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**Boundary behavior and sellers' performance in business
relationships: A contingency approach**

Leuthesser, Lance Edward, Ph.D.

The University of Texas at Austin, 1991

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Ann Arbor, MI 48106**

**BOUNDARY BEHAVIOR AND SELLERS' PERFORMANCE IN BUSINESS
RELATIONSHIPS: A CONTINGENCY APPROACH**

by

Lance Edward Leuthesser, B.S., M.B.A.

DISSERTATION

Presented to the Faculty of the Graduate School of Business

The University of Texas at Austin

in Partial Fulfillment

of the Requirements

for the Degree of

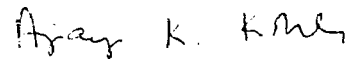
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THE UNIVERSITY OF TEXAS AT AUSTIN

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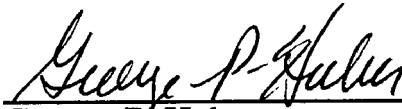
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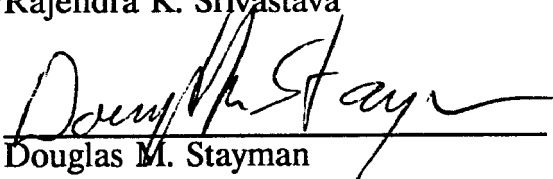
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**BOUNDARY BEHAVIOR AND SELLERS' PERFORMANCE IN BUSINESS
RELATIONSHIPS: A CONTINGENCY APPROACH**

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Lance Edward Leuthesser, Ph.D.
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Prompted by the relationship management literature which has appeared recently in marketing, we posit that the performance of sellers in ongoing buyer-seller relationships is influenced significantly by the manner in which sellers interact with buyers, which we refer to as "boundary behavior." Although a growing literature has drawn attention to the importance of relationship management, no research has investigated the extent to which, if any, relationship management leads to improved performance in terms of "hard" measures, such as sales. We fill this gap by examining how managerially controllable boundary behaviors influence sellers' sales performance. We further examine how situational factors moderate these effects, and consequently illuminate contexts in which the actions of boundary spanning personnel are most critical. The findings are likely to be of interest to managers who operate at the organization boundary, and who are therefore concerned with the effective deployment of these important resources.

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I. INTRODUCTION

The purpose of this study is to examine how the manner in which sellers interact with buyers, in ongoing industrial buyer-seller relationships (hereafter "business relationships"), influences sellers' performance. The study is motivated by the basic premise that the success of business relationships is influenced by both the product exchanged and the nature of the interactions (hereafter "boundary behavior") that take place between buyers and sellers.

This premise is supported by research which has appeared recently in the marketing literature which underscores the importance of cooperation and reciprocity in ongoing relationships (e.g. Anderson and Narus 1990; Anderson and Weitz 1989; Arndt 1979; Dwyer, Schurr and Oh 1987; Levitt 1983). Presumably, improved relationships lead to improved performance of sellers in terms of "hard" measures such as increased share of buyers' business, but no empirical research has addressed this question.

In this study we seek to address this void by assessing the influence of managerially controllable boundary behaviors on sellers' performance across a variety of business relationships. Further, we examine how contextual factors (environment, buying situation, and age of relationship) moderate this relationship. The implications of such moderating effects are particularly important

because boundary behavior involves an investment of human resources. To effectively allocate these limited resources it is essential that managers know under what circumstances the boundary behavior --> performance linkage is more, or less, strong. The main objective of this study is, therefore, to address this question.

In developing our model of seller performance, we draw upon extant theory and empirical evidence in the areas of (1) marketing channels and industrial buyer behavior, (2) negotiation behavior, (3) group behavior, and (4) organizational information processing to identify boundary behaviors which are likely to lead to enhanced performance of sellers. These behaviors relate primarily to how effectively boundary activities are coordinated, and how effective these behaviors are in creating favorable perceptions among organization members. We further identify characteristics of the environment and buying situation which are expected to moderate these effects, principally through their influence on decision-making risk. Finally, we examine the moderating effects of age of the relationship.

The model is tested by surveying buyers in a broad cross-section of business relationships. Organization informants evaluate a particular supplier ("target supplier"), concerning a particular product ("target product") for which their firm regularly uses alternative suppliers. Accordingly, assessments of target suppliers'

performance are in relation to competitive suppliers. So as to enhance the generalizability of results, a sample which is heterogeneous with respect to industry, organization size, and organization structure is used.

The next section of the paper provides the conceptual framework which underlies our model (Figure III-1). Section III discusses the variables and research hypotheses to be tested. In Section IV, measures, data collection, and data analysis are discussed. Section V is a discussion of the findings of the study. The dissertation concludes with managerial implications and directions for future research.

II. CONCEPTUALIZING BUSINESS RELATIONSHIPS

A. Defining Performance in Business Relationships

In keeping with tradition in marketing, we define performance in business relationships from the perspective of the seller. We examine two quantitative and one qualitative measure of sellers' performance. The first quantitative measure is the share of business suppliers derive from buyers, taking into consideration the number of competitive suppliers used. The second quantitative measure consists of buyers' estimates of their allocations to target suppliers, assuming such allocations were unconstrained by contractual agreements, supply limitations, switching costs, and so on. This measure is intended to draw out buyers' "idealistic" purchases which may not be reflected by the current status.

In addition to these quantitative measures, we also assess buyers' trust in sellers. Trust is regarded as a critical attitude in ongoing relationships because it captures the degree to which one party is willing to rely on another, knowing it will incur losses if the trusted party fails (Deutsch 1958; Rotter 1971; Schurr and Ozanne 1985). Buyers do not normally "hedge" by placing simultaneous orders with multiple sellers; they trust that sellers will deliver as promised. Where trust in a seller is high relative to competitive sellers, a buyer is likely to rely more on that seller (i.e., place more business with that seller). We therefore

expect that trust will be highly correlated with our quantitative measures.

B. Relationships as Emergent Groups

Organizations are related through exchanges (boundary behaviors) which occur among a subset of members (boundary spanners) belonging to both the buying and selling organizations. As a result of their ongoing interaction, these boundary spanners also comprise a separate group (Schopler 1987; Van de Ven 1976) or "social action system" (Van de Ven 1976) which is the conduit of information and resource exchanges between the two organizations (hereafter this group is referred to as the "boundary group"). In a very real sense the boundary group is the relationship. Johnston and Bonoma (1981), in a related vein, developed this notion in connection with the buying center:

The buying center exists as a communication network that does not necessarily derive its configuration nor operation from the formal organization, but rather from the regularized patterns of communication that reflect the individuals involved and their relationships (p. 146).

Although the emergent group concept has been recognized by previous researchers investigating interorganizational relationships (Schopler 1987; Van de Ven 1976), its major implication, that our understanding of such relationships

can be enhanced by examining group processes, has not been reflected in empirical research. We therefore draw, in part, upon group theory to develop our investigation of boundary behaviors, recognizing that it is the conjunction of these behaviors as both organization spanning and group actions that is a unique characteristic of interorganizational relationships.

In research on group behavior, group effectiveness is usually defined in accordance with two broad sets of criteria. The first is the effectiveness with which groups accomplish external tasks (Johnson and Johnson 1987; Shaw 1981). For boundary groups, the overriding external task involves consummating economic, i.e., product, exchanges. Outcomes of such exchanges are evaluated from both buyers' and sellers' perspectives. Specifically, buyers' assessments are expected to follow principally from their evaluations of sellers' product offerings. We acknowledge the primary importance of the offering, but in this study we wish to examine how boundary behavior affects performance after accounting for the value afforded by the offering. Accordingly, we include as a covariate the buyers' evaluation of seller's offerings.

The second set of group effectiveness criteria includes the cohesiveness and adaptiveness with which group members work to achieve external tasks, i.e., how well groups deploy their human resources independent of external task (Bradford 1964; Egan 1976; Martin 1983). This type of effectiveness is important in

predicting how well groups are likely to perform when faced with tasks which are not routine, practiced, or highly procedurized. Business relationships are characterized by varying levels of future uncertainty, and thus boundary groups require varying levels of cohesiveness and adaptiveness. We expect buyers to adjust their dependence on sellers (i.e., allocation of business to those sellers) partly in accordance with their perceptions of how effectively the boundary group is likely to deal with future contingencies. For buyers this is a simple matter of risk balancing; the greater the perceived risk associated with a particular seller, the less the dependence on that seller. Accordingly, it is this latter concept of group effectiveness (relating to cohesiveness and adaptiveness) which is our focus, and which motivates our investigation of boundary behavior as an antecedent to performance.

Activities which promote group cohesiveness and adaptiveness are frequently referred to as group maintenance functions (Ancona and Caldwell 1988; Dimock 1986). Maintenance functions relate to interaction among boundary group members (Van de Ven 1976), that is, the flow of information which influences their knowledge and perceptions. Much of what we know about group processes has been gained through the "experiential learning" approach to group research, wherein participant-observers have learned effective group skills in an environment in which external goals are de-emphasized (Bradford 1964; Gibb 1964).

Group researchers frequently identify group structure as an explanatory factor for group effectiveness (Dimock 1986; Johnson and Johnson 1987). Group structure is frequently defined in terms of the pattern of interaction among group members (Dimock 1986; Johnson and Johnson 1987). Generally, group researchers have found that when interaction among group members is extensive, that is, when all (or nearly all) members are engaged in meaningful interactions with each other, group cohesiveness and adaptiveness is enhanced (Bradford 1964; Egan 1976; Gibb 1964). Based upon findings from group research, we include boundary group structure variables among the explanatory variables in our model of business relationships (Figure III-1).

C. Boundary Group as Coordination Mechanism

A broad range of literature, including negotiation theory (Pruitt 1981), transaction cost theory (Williamson 1979), and the "relationship" literature in marketing (Anderson and Narus 1984, 1990; Dwyer, Schurr and Oh 1987; Frazier, Spekman and O'Neal 1988), converges on the notion of coordinative behavior (also referred to as "collaborative," or "cooperative" behavior) as a central desideratum of ongoing relationships. Coordinative behavior is a very general phrase referring to parties' commitment to, and behaviors directed toward, achieving mutually rewarding outcomes (Dwyer, Schurr and Oh 1987;

Pruitt 1981; Van de Ven and Walker 1984). This is in contrast with competitive behavior, wherein parties perceive that achievement of their own goals comes at the expense of their partners' (i.e., outcomes are zero sum). Over and above specific actions taken to achieve coordination in the narrow, logistical sense, coordinative behavior includes actions which show that partners' are committed to achieving mutual benefit. Coordinative interorganizational behavior and cohesive group behavior are achieved simultaneously in the boundary group.

Business relationships are seldom circumscribed by detailed contracts which spell out responses to every contingency, and most managers are averse to pursuing such contracts (Macaulay 1963). Boundary group members, as boundary spanners, may be expected to experience relatively high levels of role conflict and role ambiguity as they strive to reconcile differences in organizational goals which inevitably arise (Lyonski 1985). Therefore, for relationship stability, coordinative behavior is necessary from two viewpoints: (1) integrating organizational goals, and (2) promoting the psychological well-being of boundary group members, who in turn influence the perceptions and attitudes of other organization members. Anderson and Narus (1990) found coordination to have a positive influence on trust in channel relationships, and research on industrial buyer behavior has shown that industrial buyers strongly prefer to use a collaborative problem-solving approach in dealing with sellers (Day, Michaels and

Perdue 1988; Dion and Banting 1988; Perdue, Day and Michaels 1986). No research, however, has investigated the influence of coordinative behavior on sales-related performance.

Specific coordinative behaviors relate to the manner in which boundary group members provide information about themselves and their organizations, as well as the manner in which they acquire information about their partners (Pruitt 1981). We distinguish coordinative behavior from coordinative mechanisms, such as formal rules and procedures (Ruekert and Walker 1987), on the basis of the former's ability to convey motives and intentions in addition to providing specific information, or heuristics, for coordinating activities. Sellers' coordinative behaviors comprise the second set of explanatory variables in Figure III-1.

D. Boundary Spanning

In order to effectively coordinate activities, related organizations must process information across their respective boundaries. Accordingly, boundary group effectiveness is likely to be influenced by its capacity as an information processing unit. Daft and Lengel (1986) suggest that organizations process information for two reasons: (1) to reduce uncertainty, and (2) to reduce equivocality (ambiguity). Uncertainty means the absence of information, and

corresponds to the amount of information that must be processed, whereas equivocality refers to the existence of multiple or conflicting interpretations, and relates to the ability of the information channel to deal with complex information and/or subtle differences in meaning.

We assume organizations have both information processing requirements, and accordingly define information capacity of the boundary group in terms of its ability to accommodate both kinds of information processing needs. The third set of explanatory variables in Figure III-1 relates to the information capacity of the boundary group.

E. Context Effects

The boundary group approach to business relationships is "micro," when viewed from the perspective of the organization as a whole. From the buyer's perspective, a seller represents but one element in a larger environment. The boundary group perspective focuses on interpersonal processes, independent of larger organizational considerations. Assuming interpersonal processes are influential in establishing boundary group norms and behaviors, then the evolution of boundary groups may not necessarily "fit" the needs of the organization as viewed from a larger perspective. For example, buyers facing rapid technological change or other dynamic elements in their output market

generally require more adaptive, responsive business relationships than do buyers facing relatively stable output markets (Ruekert and Walker 1987; Walker and Ruekert 1987). But what will result if the boundary group is not adaptive or responsive?

Stated differently, the foregoing question asks to what extent, if any, the influence of boundary behaviors on sellers' performance is contingent on the context of the business relationship. In addition to buyers' output market environment, characteristics of the product may also be expected to moderate the boundary behavior --> performance linkage because, from the buyer's point of view, different buying situations may be more, or less, critical to buyers' overall profitability. Finally, we expect age of the relationship to have a moderating effect, because over time buyers and sellers tend to form enduring beliefs which are not necessarily related to their current interaction (Anderson and Weitz 1989). Figure III-1 therefore incorporates the moderating influence of buyers' environment, product characteristics, and age of relationship.

F. Domain of Study

Researchers emphasize that voluntary exchanges between organizations may be placed along a continuum ranging from "discrete" ("market") on one end to "relational" on the other end, depending on the parties' level of incentive for, or

commitment to, an ongoing relationship (Dwyer, Schurr and Oh 1987; Macneil 1980; Ruekert, Walker and Roering 1985; Williamson 1979). It is only in instances in which exchange is at least moderately "relational" that our framework is relevant. Williamson (1979) conceptualizes such relationships as those requiring investment of significant idiosyncratic human capital, in the form of relationship specific training and learning-by-doing. We place this study in the relational domain by examining only business relationships which (1) have been ongoing for one year or longer, and (2) involve a product, in terms of potential impact on buyers' profitability, of at least moderate importance. We therefore expect at least a moderate level of relationship-specific human capital in our sample of business relationships.

G. Summary

We define sellers' performance in business relationships in terms of the share of business sellers obtain (or might obtain) from buyers, as well as buyers' trust in them. Traditionally, marketers have focused on the marketing mix as the principal factor explaining sellers' performance. Recent marketing literature has emphasized the importance of the "human" element in business relationships, which we view from the perspective of the boundary group.

Boundary groups emerge as a result of ongoing interaction, and form the

primary mechanism of coordination in business relationships. Coordination entails the processing of information, not only for specific, logistical purposes, but also as the basis for boundary group members' perceptions and motivational inferences. Three sets of boundary behavior variables are expected to influence sellers' performance: (1) boundary group structure, (2) sellers' coordinative behaviors, and (3) boundary information capacity. Boundary group structure refers to the interaction pattern in the boundary group. Sellers' coordinative behavior refers to sellers' information giving and information seeking style. Boundary information capacity refers to the amount and complexity of information which the boundary group is able to process.

The environment, buying situation, and age of the relationship are expected to moderate the influence of boundary behavior on sellers' performance through their effects on decision-making uncertainty, decision-making importance, and belief formation, respectively.

III. MODEL OF SELLERS' PERFORMANCE

A. Dependent Variables

1. Share Ratio - Actual. Actual share ratio refers to the ratio of sellers' actual share of buyers' business divided by their "expected" share, where expected share is the inverse of the number of suppliers used by buyers' for the focal product. A share ratio of greater than 1.0 thus indicates a share premium, whereas a ratio of less than 1.0 indicates a share penalty.

Buyers are expected to spread their business among a set of suppliers such that their dependence on each represents a balanced risk. Buyers' perceptions of risk associated with sellers are expected to follow from their perceptions of the (1) uncertainty, and (2) significance of the outcomes from dealing with those sellers (March and Shapira 1987). Where uncertainty is lower, the significance of outcomes must therefore be greater for a given level of perceived risk. The more effectively coordinated the business relationship, the less uncertainty buyers will associate with an inherently unpredictable future; buyers will be confident that they will be able to adapt. Factors such as price, quality, delivery, and service being equal, sellers in better coordinated relationships are expected to receive more of buyers' business.

2. Share Ratio - Predicted. Predicted share ratio is based on buyers' estimates of business that sellers would receive if allocations were based solely on sellers' performance. We include predicted share ratio as a dependent measure to assess buyers' intentions which, for reasons unrelated to performance evaluations (e.g., binding contracts, switching costs, production limitations), may not be reflected in actual share. As modeled in Figure III-1, predicted share ratio and actual share ratio are expected to be influenced similarly, and the subsequent discussion addresses these two variables collectively as "share."

3. Trust. Trust refers to the degree to which buyers believe sellers are reliable and may be counted on to act in buyers' interests (Schurr and Ozanne 1985). As has been noted in the literature, trust implies a willingness to put oneself at risk, because there is always the possibility that the trusted party may not act in one's interests (Deutsch 1958; Rotter 1971; Rempel, Holmes and Zanna 1985). Trust is strongly linked to ongoing exchange because it is not possible to specify in advance all possible contingencies which can affect a relationship. One party in a relationship must inevitably act on the expectation that the other will respond as desired, and thus must trust the other (Blau 1974). Trust has been emphasized in recent marketing channels literature as an

important factor in relationship continuity (Anderson and Weitz 1989; Anderson and Narus 1990).

B. Explanatory Variables

1. Seller's Vertical Involvement. Seller's vertical involvement refers to the degree to which interaction with the buyer is hierarchically dispersed in the seller's organization. Boundary group members, acting as representatives for the organization as a whole, are accountable to others in the hierarchy (upward and downward) for their actions as organization representatives. Where sellers' hierarchical representation is greater, buyers will perceive less seller accountability outside the boundary group, and consequently less likelihood of agreements with boundary personnel being modified, or countermanded (Pruitt 1981). Where the perceived risk of modification or abrogation of agreements is less, buyers will associate less uncertainty with boundary group decisions, and will be more likely to increase their dependence on sellers by increasing share. We thus hypothesize that seller's vertical involvement will be positively associated with share.

Seller's vertical involvement is also expected to be positively related to buyer's trust. Group researchers have consistently reported that the level of trust in groups is influenced positively by the extent to which all group members are

"connected," i.e., all engaged in ongoing interaction (Bradford 1964; Gibb 1964; Golembiewski and McConkie 1975; Martin 1983). The distinction between participants and observers in groups actualizes this concept. An overarching group objective is to achieve a condition in which all members are participants (Bradford 1964). Observers lower the level of group trust because it is uncertain whether or not they have "bought in" to the group norms and commitments, and thus it is unclear whether they are motivated to support group decisions (with which they may not fully agree).

The boundary group is embedded in a larger system comprising the respective parent organizations. From the buyer's perspective, there are "observers" in the seller organization to the extent that the boundary group fails to subsume it entirely ("complete" trust may be viewed as a goal, not fully attainable in the small group sense). Seller's vertical involvement is one measure of the degree to which the boundary group subsumes seller organizations, and is therefore expected to correlate positively with the perceived level of "participants" in the seller organizations. We therefore hypothesize that seller's vertical involvement will be positively related to buyer's trust.

2. Seller's Lateral Involvement. Seller's lateral involvement refers to the degree to which interaction with the buyer is functionally dispersed within the

seller organization. Business relationships typically require the coordination of activities involving a number of functional areas (Walker and Ruekert 1987). The more extensively representatives of these functional areas are involved in interaction with the buyer, the greater is the expected level of participative coordination (McCann and Galbraith 1981; Walker and Ruekert 1987).

Participative coordination refers to mutual resolution of conflict, or potential conflict, among the parties whose activities must be coordinated. This is in contrast to hierarchical coordination, which refers to imposition of higher authority, usually through required adherence to formal rules or operating procedures (Walker and Ruekert 1987). Participative coordination is expected to lead to fuller understanding of, and more adaptive solutions to, problems which may cut across functional areas. Effective coordination on the part of the seller will result in fewer problems adversely affecting the buyer, and should result in greater perceived competence and enhanced confidence in sellers (i.e., lower perceived risk associated with those sellers). Other factors being equal, buyers are more likely to shift business toward sellers in which they perceive less risk. Accordingly, we hypothesize the level of sellers' lateral involvement to be positively related to share.

Following from the same argument advanced for vertical involvement, sellers' lateral involvement is expected to be positively related to the perceived level of

"participants" in the seller organization. Where the level of participants is higher, trust is expected to be correspondingly higher. We therefore hypothesize that sellers' lateral involvement will be positively related to buyers' trust.

3. Buyer's Vertical and Lateral Involvement. Buyer's vertical involvement refers to the degree to which interaction with the seller is hierarchically dispersed within buyer's organization. Buyer's lateral involvement refers to the degree to which interaction with the seller is functionally dispersed within the buyer's organization. Buyers' vertical involvement is expected to relate primarily to the level of approval of, and organizational commitment to, boundary group agreements, whereas lateral involvement is expected to relate primarily to coordination of functionally interdependent activities.

More extensive vertical involvement within the buyer organization will result in greater opportunity for responsible members to directly negotiate with the seller regarding scope of activities, mutual responsibilities, terms and conditions of payment, remedies and recourse, etc. More extensive lateral involvement will result in greater opportunity to coordinate functional assignments, set schedules, establish liaison, etc. We expect both of these measures to be positively related to the level of coordinative effort, and we assume that, on average, such efforts are effective. Accordingly, both variables are expected to exert a direct influence

on coordination effectiveness.

Further, owing to bounded rationality, individual boundary group members within buyers' organizations are not likely to have familiarity with all details of the business relationship. Where these individuals perceive extensive boundary group involvement, therefore, they are likely to conclude that others in their organization have "signed off" in their respective areas. Consequently, they are likely to have greater generalized confidence in dealings with that seller, even pertaining to areas outside of their direct responsibility. Therefore, in addition to their direct impact on coordination effectiveness in the logistical sense, greater vertical and lateral involvement also operate through their influence on the perceptions of boundary group members.

In sum, greater vertical and lateral involvement of buyers is expected to lead to (1) more effective coordination with sellers, as well as (2) perceptions which enhance buyers' general level of confidence in sellers. Accordingly, we hypothesize that buyers' vertical involvement and buyers' lateral involvement will positively influence share.

Buyers' vertical involvement and lateral involvement are also expected to positively influence buyers' trust. As buyers' involvement with sellers becomes more extensive, greater opportunities are available for members of seller organizations to demonstrate trustworthy behavior. In the process of interaction

members of both organizations are able to learn the needs and expectations of the other, and communicate their respective intentions (Egan 1976). In everyday circumstances, numerous instances occur in which boundary group members can display trust. Seemingly insignificant interactions such as returning phone calls promptly, anticipating and responding to others' concerns without being asked, and making the effort to confirm commitments, all build trust (Golembiewski and McConkie 1975). Based on the assumption that exchanges are, on average, used propitiously to this end, greater interaction will enhance the trust of both parties. Empirical support for this conclusion has been found in marketing channels, where greater channel member participation in decision making was found to positively influence trust (Dwyer and Oh 1987).

Following from the above discussion, the following hypotheses are offered regarding the influence of boundary group structure on performance:

- H1A: Sellers' vertical involvement will have a positive influence on actual share ratio.
- H1B: Sellers' lateral involvement will have a positive influence on predicted share ratio.
- H1C: Buyers' vertical involvement will have a positive influence on actual share ratio.
- H1D: Buyers' lateral involvement will have a positive influence on actual share ratio.

H2A: Sellers' vertical involvement will have a positive influence on predicted share ratio.

H2B: Sellers' lateral involvement will have a positive influence on predicted share ratio.

H2C: Buyers' vertical involvement will have a positive influence on predicted share ratio.

H2D: Buyers' lateral involvement will have a positive influence on predicted share ratio.

H3A: Sellers' vertical involvement will have a positive influence on trust.

H3B: Sellers' lateral involvement will have a positive influence on trust.

H3C: Buyers' vertical involvement will have a positive influence on trust.

H3D: Buyers' lateral involvement will have a positive influence on trust.

4. Disclosing. Disclosing refers to the extent, as perceived by buyers, to which sellers provide information regarding their (sellers') plans, programs, and objectives. In business relationships it is imperative that each party have a sufficient understanding of the other's business so as to permit adaptive problem solving. In support of this, Anderson and Weitz (1989) have found that when channel members keep their partners well informed as to the details of their

business, partners are likely to place greater confidence in them, coordinate more effectively with them, and be more willing to rely on them.

Business relationships involve a process of ongoing negotiation, that is, a continual resolution, or prevention, of conflict. Pruitt (1981) stresses the importance of information exchange in achieving coordinative agreements. In simulation of buyer-seller bargaining Pruitt and his associates have found that providing explicit information about goals, values, and priorities consistently leads to increased levels of "integrative agreements" (Carnevale, Pruitt and Seilheimer 1981; Pruitt and Lewis 1975; Pruitt and Lewis 1977). Integrative agreements are those which entail a high level of mutual satisfaction with the outcomes. Greater disclosing by the seller is expected to lead to more effective coordination because it provides buyers with a rich data base for the formulation of adaptive solutions to conflict which are perceived to be based on the genuine needs of the seller (Pruitt 1981). Because disclosing has been shown to lead to coordination effectiveness, we hypothesize that disclosing will have a positive influence on share.

In small groups, disclosing is risk-taking behavior because it opens the discloser to possible rebuff or ridicule (Egan 1976; Golembiewski and McConkie 1975; Martin 1983). Analogously, disclosing in business relationships may entail giving information which is potentially damaging to the seller. Examples of such

disclosures would be the revealing of plans and goals, which if passed on to competitors could be used to sellers' detriment. When such disclosing is done because it is in the best interests of the relationship, it is perceived as a trusting action on the part of sellers, and should in turn enhance buyers' trust. In support, Anderson and Weitz (1989) found that the level of communications concerning plans, programs, expectations, goal setting, and performance evaluation was positively related to trust in channel relationships. We therefore hypothesize that disclosing will have a positive influence on trust.

5. Initiating. Initiating refers to the degree to which sellers overtly seek relevant information about buyers' business operations and market needs. Initiating is a relationship-specific form of intelligence gathering. If all relevant and important information needed to effectively coordinate activities in a business relationship were voluntarily offered, then initiating would not be necessary. However, owing to bounded rationality, buyers are unlikely to know all of the information which is either useful, or necessary, for sellers' optimal fulfillment of their obligations in the business relationship. The degree to which sellers actively seek this information is likely to be predictive of the degree to which they actually possess such information. The degree to which sellers possess needed information is, in turn, likely to positively influence coordination

effectiveness, and share.

Initiating behavior is also expected to positively influence buyers' perceptions of sellers' motivation to perform in their (buyers') interests. Where sellers show high initiation, they are likely to be perceived as "intrinsically" motivated (i.e., motivated by concerns for the mutual welfare of the parties), and consequently trusted more by buyers (Rempel, Holmes and Zanna 1985).

6. Signaling. Signaling refers to the degree to which sellers communicate their intended actions in the relationship. Signaling by sellers is the most direct form of coordinative behavior because it (1) clarifies/confirms planned actions, and (2) provides an opportunity for buyers to modify planned actions before they are (potentially) erroneously enacted. Accordingly, we expect signaling to positively influence coordination effectiveness, and share.

Signaling also complements initiating by providing feedback on information gained from buyers. In small group settings this has been described as "reflecting accurate empathy" (Egan 1976), or "clear signaling" (Golembiewski and McConkie 1975). Signaling closes the loop begun with initiating. Signaling presents an opportunity for sellers to confirm that they have attended to sellers' needs and expectations, and that those needs and expectations will be responded to. In relationships, partners must not only be empathetic with their counter-

parts, they must actively convey this (Egan 1976). In group research, signaling has been shown to be a trust-building behavior (Egan 1976; Golembiewski and McConkie 1975); accordingly we hypothesize that it will positively influence buyers' trust in business relationships.

Based upon the foregoing discussion, we offer the following hypotheses regarding the influence of sellers' coordinative behavior on performance:

H4A: Disclosing will have a positive influence on actual share ratio.

H4B: Initiating will have a positive influence on actual share ratio.

H4C: Signaling will have a positive influence on actual share ratio.

H5A: Disclosing will have a positive influence on predicted share ratio.

H5B: Initiating will have a positive influence on predicted share ratio.

H5C: Signaling will have a positive influence on predicted share ratio.

H6A: Disclosing will have a positive influence on trust.

H6B: Initiating will have a positive influence on trust.

H6C: Signaling will have a positive influence on trust.

7. Frequency. Frequency is defined as the inverse of the average time between interactions between buyer and seller. Other factors being equal, frequency relates positively to the amount of information processed in the boundary group, and to the ability of boundary interaction to reduce uncertainty (Daft and Lengel 1986). On balance, greater uncertainty reduction should lead to greater coordination effectiveness (Anderson and Weitz 1989), and greater share.

The finding of a positive relationship between communication frequency and trust has been fairly robust, both within the marketing literature (Anderson and Narus 1990; Anderson and Weitz 1989) and the literature on group behavior (Bradford 1964; Loomis 1959; Golembiewski and McConkie 1975). In general, the more frequently individuals communicate, the more they are perceived by their partners to be motivated in the interest of the relationship, and the more they are trusted. Accordingly, we hypothesize that frequency will have a positive influence on buyers' trust.

8. Richness. Richness is defined as the proportion of interactions between the buyer and seller which are conducted face-to-face. Face-to-face communication is a richer medium than impersonal modes (Daft and Lengel 1986). In contrast to quantity of information, richness refers to the ability to process

ambiguous information (Daft and Lengel 1986). Rich communication media (e.g., face-to-face meetings) are most effective in dealing with ambiguity. Organizations are expected to have both types of information processing needs (Daft and Lengel 1986; Huber and Daft 1987), and we assume both types of information must be processed for effective coordination in business relationships. We further assume that, due to considerations of cost and logistics, most business relationships are characterized by sub-optimal levels of face-to-face communication (sub-optimal in an information processing sense, not cost/benefit sense). Accordingly, increasing the proportion of face-to-face communication is expected to produce a more balanced fit with the actual information load, and enhance coordination effectiveness. We therefore hypothesize that richness will positively influence share.

Richness is expected to have a positive influence on buyers' trust because face-to-face interaction has the capacity to convey non-verbal cues and voice inflections which can substantially influence individuals' perceptions of others (Johnson and Johnson 1987; Shaw 1981). Indeed, some researchers consider face-to-face interaction to be a necessary condition for group development (Bonner 1959; Homans 1950). Considering the extent to which face-to-face interaction and non-verbal information processing has been emphasized in the trust literature, we expect richness to have a positive influence on buyers' trust.

Based on the foregoing discussion of frequency and richness, the following hypotheses are advanced:

H7A: Frequency will have a positive influence on actual share ratio.

H7B: Richness will have a positive influence on actual share ratio.

H8A: Frequency will have a positive influence on predicted share ratio.

H8B: Richness will have a positive influence on predicted share ratio.

H9A: Frequency will have a positive influence on trust.

H9B: Richness will have a positive influence on trust.

C. Moderator Variables

The hypothesized moderating effects for each moderator variable - dependent variable combination embodied in Figure III-1 are discussed below.

1. Environmental Dynamism - Share. Environmental dynamism refers to the perceived frequency of change in market forces in buyers' environments (Achrol

and Stern 1988). Environmental dynamism has been found to be positively related to decision-making uncertainty in marketing channels (Achrol and Stern 1988). This finding is in general agreement with the literature which has investigated the effects of environmental uncertainty (Duncan 1972; Huber and Daft 1987; Leblebici and Salancik 1981; Thompson 1967).

Where decision-making uncertainty is high, the relative importance of participative coordination is expected to be greater than when decision-making uncertainty is low, because more flexible and innovative approaches to the resolution of interdepartmental conflict are likely to be required (Walker and Ruekert 1987). Therefore, the greater the environmental dynamism, the more important the coordination enhancing effects of sellers' and buyers' lateral involvement are expected to be.

Sellers' and buyers' vertical involvement both relate to the dispersion of decision-making authority represented in the boundary group. We expect the responsiveness of the boundary group as a decision-making unit to be enhanced by greater vertical involvement, because accountability to members outside the boundary group is lessened (more key decision makers belong to the group). Where environmental dynamism is high, the responsiveness of the boundary group is likely to be a more important consideration than when environmental dynamism is low, because the boundary group will be called upon to make more

decisions, more quickly (Huber and Daft 1987; Walker and Ruekert 1987). Accordingly, we hypothesize a greater influence of vertical involvement (buyers' and sellers') when environmental dynamism is high.

Sellers' coordinative behaviors are also expected to have a stronger influence on performance under conditions of high environmental dynamism. Where decision-making uncertainty is high, the need to make more innovative decisions, and consequently the need for more extensive knowledge of sellers, is greater than under conditions of low decision-making uncertainty (Walker and Ruekert 1987). Sellers' disclosing gives buyers' a basis from which to formulate more adaptive solutions to problems, and is therefore expected to be more instrumental in achieving coordination effectiveness under conditions of high, versus low, decision-making uncertainty. Likewise, the coordination enhancing effects of sellers' initiating and signaling should have a greater impact when decision-making uncertainty is high.

Environmental uncertainty has typically been examined as an antecedent of boundary spanning activity (Katz and Kahn 1966; Schwab, Ungson and Brown 1985; Thompson 1967), although there is evidence that organization structure may influence boundary spanning more than environmental uncertainty (Leifer and Huber 1977). However, no research has investigated the moderating effect of environmental uncertainty on boundary spanning as a predictor of perfor-

mance. Greater environmental dynamism is expected to result in a greater amount of product-related information for processing in the boundary group (Ruekert and Walker 1987; Walker and Ruekert 1987). Frequency alone cannot completely capture the amount of information processed, which is related to both the frequency and duration of communication (Farace, Monge and Russell 1977). Other factors being equal, however, greater communication frequency will correspond to a greater amount of information being processed. On average, the greater the level of uncertainty, the greater the uncertainty reduction per exchange (exchanges will tend to be longer or more dense), and the greater the decision enhancing capacity of each exchange.

Environmental dynamism increases primarily the amount of information which must be processed in the boundary group. Accordingly, we expect that as the amount of information to be processed increases, other factors being equal, communication modes other than face-to-face are likely to be more effective, because they tend to be more timely. Based on the foregoing discussion, we hypothesize:

The greater the environmental dynamism, the stronger (weaker)
the relationship between actual share ratio and:

- H10A: Sellers' vertical involvement.
- H10B: Sellers' lateral involvement.
- H10C: Buyers' vertical involvement.

H10D: Buyers' lateral involvement.
 H10E: Disclosing.
 H10F: Initiating.
 H10G: Signaling.
 H10H: Frequency.
 H10I: (Richness).

The greater the environmental dynamism, the stronger (weaker)
 the relationship between predicted share ratio and:

H11A: Sellers' vertical involvement.
 H11B: Sellers' lateral involvement.
 H11C: Buyers' vertical involvement.
 H11D: Buyers' lateral involvement.
 H11E: Disclosing.
 H11F: Initiating.
 H11G: Signaling.
 H11H: Frequency.
 H11I: (Richness).

2. Environmental Dynamism - Trust. In dynamic environments, where decision-making uncertainty is greater, sellers' coordinative behaviors are expected to be more strongly related to trust than when environmental dynamism is low. Under conditions of high uncertainty the risk of making poor decisions is greater and it is therefore more important that each party fully understand the capabilities, limitations, goals, needs, and intentions of the other. Viewing boundary behavior as a form of risk-taking behavior, sellers' disclosing, initiating, and signaling are expected to have greater influence on buyers' trust because

these behaviors carry more "psychological weight" when decision-making risk is high (Golembiewski and McConkie 1975). Support for this conclusion can be found in group research, where trust has been found to be enhanced more when greater risk-taking behavior is displayed (Gibb 1964). Accordingly, we hypothesize:

The greater the environmental dynamism, the stronger the relationship between trust and:

H12A: Disclosing.

H12B: Initiating.

H12C: Signaling.

3. Environmental Diversity - Share. Environmental diversity refers to the degree of heterogeneity in buyers' output markets for the focal product. Where buyers' customers are diverse with respect to demographics, industry, and preferences for product features, buyers' strategic behavior is likely to approximate "prospector" businesses (Miles and Snow 1978; Walker and Ruekert 1987). Under such conditions, greater interfunctional conflict and need for coordination (within buyer organizations) is expected because managers must contend with, and adapt to, a broader array of unfamiliar situations (Walker and Ruekert 1987). On average, this adaptation is likely to also complicate the coordination task on the part of sellers, as they are called upon to make modifications in their

offerings to meet the diverse market needs of buyers. Dwyer and Welsh (1985) argue that, due to such added complexities, heterogeneous environments in channels require more decentralized decision structures and more participative decision making. Accordingly, we hypothesize that the coordination enhancing effects of sellers' lateral involvement and buyers' lateral involvement will be relatively more important under conditions of high environmental diversity than under low diversity.

Greater vertical involvement in the boundary group means greater decision-making authority and less accountability, thus enabling it to respond to a broader range of situations than when vertical involvement is lower. Where buyers serve a diverse set of customers, with diverse needs (which, in turn, face a diverse set of environments), buyers will be presented with (1) more frequent decision-making situations, and (2) more varied decision-making situations. Environmental diversity is, therefore, expected to increase the importance of the decision-making responsiveness of the boundary group.

Sellers' coordinative behaviors are also expected to assume greater importance under conditions of high environmental diversity. Specifically, sellers' disclosing, initiating, and signaling will have greater impact when the number and complexity of problems are greater, and the potential adverse impact of sub-optimal solutions is greater. Under such conditions the risk reducing potential

of these behaviors is greater than when environmental diversity is lower.

Environmental diversity is expected to lead to a relatively greater increase in the ambiguity of information than uncertainty of information (Daft and Lengel 1986; Dwyer and Welsh 1985), although both information processing requirements are expected to increase. With high environmental diversity, therefore, communication frequency should have a stronger influence on coordination effectiveness because, on average, each interaction has greater potential for reducing uncertainty. Because ambiguity plays a greater role in information processing in diverse environments, organizations must "enact" solutions to problems to a greater extent than "learning" those solutions through the collection of additional data (Weick 1979). Consequently, richness is expected to have a greater influence on information processing effectiveness when environmental diversity is greater. In summary:

The greater the environmental diversity, the stronger the relationship between actual share ratio and:

- H13A: Sellers' vertical involvement.
- H13B: Sellers' lateral involvement.
- H13C: Buyers' vertical involvement.
- H13D: Buyers' lateral involvement.
- H13E: Disclosing.
- H13F: Initiating.
- H13G: Signaling.
- H13H: Frequency.
- H13I: Richness.

The greater the environmental diversity, the stronger the relationship between predicted share ratio and:

- H14A: Sellers' vertical involvement.
- H14B: Sellers' lateral involvement.
- H14C: Buyers' vertical involvement.
- H14D: Buyers' lateral involvement.
- H14E: Disclosing.
- H14F: Initiating.
- H14G: Signaling.
- H14H: Frequency.
- H14I: Richness.

4. Environmental Diversity - Trust. Since environmental diversity is expected to increase decision-making risk, the trust enhancing effects of disclosing, initiating, and signaling are expected to be more influential under conditions of high diversity than under conditions of low diversity. This conclusion follows from the expectation of greater psychological impact of coordinative communication (riskier decision context). Accordingly, disclosing useful information, showing motivation to act in partners' interests, and signaling intentions should enhance trust to a greater degree when environmental diversity is greater. Therefore:

The greater the environmental diversity, the stronger the relationship between trust and:

- H15A: Disclosing.
- H15B: Initiating.
- H15C: Signaling.

5. Product Importance - Share and Trust. Product importance refers to the perceived impact of the product on buyers' profitability. Product importance is expected to be directly related to the magnitude of decision-making outcomes, and thus decision-making risk (March and Shapira 1987). Where the economic value of the product is greater, or the product otherwise has greater potential to affect buyers' profitability (via impact on quality, productivity, etc.), buyers are likely to place higher priority on those product-related decisions, and attend to those decisions to a greater degree than for products of lesser importance.

This conclusion has been supported in buying center research, where product importance has been found to be positively related to the amount of product-related communication offered (McQuiston 1989), and positively related to both vertical and lateral involvement (Johnston and Bonoma 1981). We expect the same effects, on average, to obtain in business relationships. No research, however, has examined the moderating effect of product importance on boundary behavior as a predictor of performance.

Clearly, where product importance is greater, any variability in decision outcomes has a magnified effect compared to less important situations; buyers are therefore likely to place more weight on the quality of those decisions. Coordination and risk reduction are therefore expected to be more strongly predictive of performance in more important situations. It follows logically, then,

that where product importance is greater, the main effects posited in Figure III-1 will be stronger than under conditions of lower product importance. We therefore hypothesize that:

The greater the product importance, the stronger the relationship between actual share ratio and:

- H16A: Sellers' vertical involvement.
- H16B: Sellers' lateral involvement.
- H16C: Buyers' vertical involvement.
- H16D: Buyers' lateral involvement.
- H16E: Disclosing.
- H16F: Initiating.
- H16G: Signaling.
- H16H: Frequency.
- H16I: Richness.

The greater the product importance, the stronger the relationship between predicted share ratio and:

- H17A: Sellers' vertical involvement.
- H17B: Sellers' lateral involvement.
- H17C: Buyers' vertical involvement.
- H17D: Buyers' lateral involvement.
- H17E: Disclosing.
- H17F: Initiating.
- H17G: Signaling.
- H17H: Frequency.
- H17I: Richness.

The greater the product importance, the stronger the relationship between trust and:

- H18A: Sellers' vertical involvement.
- H18B: Sellers' lateral involvement.
- H18C: Buyers' vertical involvement.
- H18D: Buyers' lateral involvement.
- H18E: Disclosing.
- H18F: Initiating.
- H18G: Signaling.
- H18H: Frequency.
- H18I: Richness.

6. Product Type - Share and Trust. Product type refers to whether the focal product is a good, or service. The most frequently cited difference between goods and services is intangibility (Berry 1980; Lovelock 1981; Shostack 1977; Zeithaml, Parasuraman and Berry 1985). Because services cannot be seen, felt, tasted, or touched in the same manner as goods, buyers frequently find the task of evaluating service performance to be more difficult than for goods. Buyers frequently rely on cues obtained from the service context to infer the quality and performance of services (Berry 1980; Shostack 1977).

Another distinguishing characteristic of services is inseparability of production and consumption. For many services this forces the buyer into intimate contact with the production process (Carmen and Langeard 1980), and also results in marketing and production being highly interactive (Gronroos 1978). In the case of services, then, boundary behavior and the service itself are

often inseparable.

For these reasons, service buyers are expected to place greater reliance on boundary behaviors to guide their purchase behavior than are goods buyers. Other factors being equal, the main effects posited in Figure III-1 are expected to be stronger in business relationships involving services, as compared to goods. Accordingly:

For services, as compared to goods, actual share ratio will be influenced more strongly by:

- H19A: Sellers' vertical involvement.
- H19B: Sellers' lateral involvement.
- H19C: Buyers' vertical involvement.
- H19D: Buyers lateral involvement.
- H19E: Disclosing.
- H19F: Initiating.
- H19G: Signaling.
- H19H: Frequency.
- H19I: Richness.

For services, as compared to goods, predicted share ratio will be influenced more strongly by:

- H20A: Sellers' vertical involvement.
- H20B: Sellers' lateral involvement.
- H20C: Buyers' vertical involvement.
- H20D: Buyers' lateral involvement.
- H20E: Disclosing.
- H20F: Initiating.
- H20G: Signaling.
- H20H: Frequency.
- H20I: Richness.

For services, as compared to goods, trust will be influenced more strongly by:

- H21A: Sellers' vertical involvement.
- H21B: Sellers' lateral involvement.
- H21C: Buyers' vertical involvement.
- H21D: Buyers' lateral involvement.
- H21E: Disclosing.
- H21F: Initiating.
- H21G: Signaling.
- H21H: Frequency.
- H21I: Richness.

7. Age of Relationship - Trust. Trust is posited to result from evolutionary processes; deepening beliefs are formed through the accumulation of evidence about the relationship partner (Anderson and Narus 1990; Dwyer and Lagace 1986; Rempel, Holmes and Zanna 1985). The literature on trust emphasizes the importance of the early stages of relationships as a critical period for the trust-building process (Dwyer, Schurr and Oh 1987; Giffen 1967; Rempel, Holmes and Zanna 1985). In the initial stages of business relationships, therefore, boundary behaviors are expected to correspond more closely to buyers' trust than in later stages when beliefs have been relatively well established. Everyday examples of this, on an interpersonal level, would include trusted old friends or relatives with whom one had not interacted in years. Under such conditions, the relationship is in a "holding" or "maintenance" mode; absent disconfirming evidence, beliefs remain intact. A similar evolution is likely to prevail in organizational relation-

ships. In support, Anderson and Weitz (1989) found that, while there was, overall, a strong positive relationship between the level of communication and trust in marketing channel dyads, older dyads tended to be "communication efficient," apparently sustained by "inertia" (p. 320). Therefore:

The greater the age of the relationship, the weaker the relationship between trust and:

- H22A: Sellers' vertical involvement.
- H22B: Sellers' lateral involvement.
- H22C: Buyers' vertical involvement.
- H22D: Buyers' lateral involvement.
- H22E: Disclosing.
- H22F: Initiating.
- H22G: Signaling.
- H22H: Frequency.
- H22I: Richness.

For a summary of hypothesized effects on actual share ratio, predicted share ratio, and trust refer to Tables III-1, III-2 and III-3 respectively.

D. Covariates

1. Product Rating. Product rating refers to buyers' evaluations of how well sellers perform on the basic marketing mix elements of price, quality, delivery, and service. Satisfactory execution on the basic offering is the sine qua non of

any business relationship, and is expected to significantly influence share and buyers' trust. We include a composite score for product rating as a covariate in all equations modeling sellers' performance.

2. Sellers' Dependence. Sellers' dependence refers to the estimated proportion of sellers' total sales provided by buyers. Where buyers' perceive that they represent a greater share of sellers' business, they are likely also to perceive that sellers have greater dependence on them and, therefore, greater motivation to act in buyers' interests. Accordingly, buyers are expected to have greater trust in sellers. We therefore include sellers' dependence as a covariate in all equations modeling buyers' trust.

FIGURE III-1

Hypothesized Model of Sellers' Performance in Business Relationships

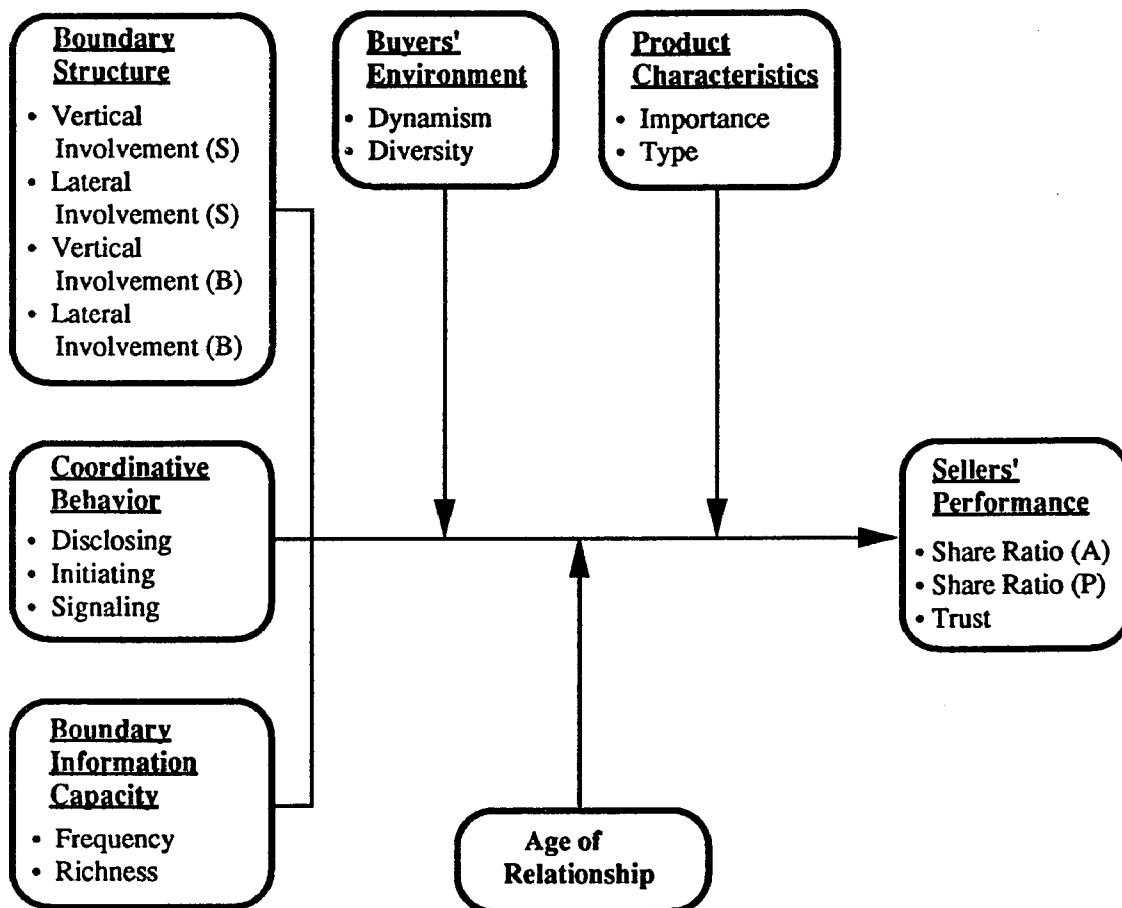


TABLE III-1

Hypothesized Effects - Dependent Variable Actual Share Ratio

Explanatory Variable	Main Effect	Moderating Effects			
		Dynamism	Diversity	Importance	Type
Sellers' vertical involvement	+ ^a (1A)	> ^b (10A)	> (13A)	> (16A)	> (19A)
Sellers' lateral involvement	+ (1B)	> (10B)	> (13B)	> (16B)	> (19B)
Buyers' vertical involvement	+ (1C)	> (10C)	> (13C)	> (16C)	> (19C)
Buyers' lateral involvement	+ (1D)	> (10D)	> (13D)	> (16D)	> (19D)
Disclosing	+ (4A)	> (10E)	> (13E)	> (16E)	> (19E)
Initiating	+ (4B)	> (10F)	> (13F)	> (16F)	> (19F)
Signaling	+ (4C)	> (10G)	> (13G)	> (16G)	> (19G)
Frequency	+ (7A)	> (10H)	> (13H)	> (16H)	> (19H)
Richness	+ (7B)	< ^c (10I)	> (13I)	> (16I)	> (19I)

^a To be read as: Sellers' vertical involvement will have a positive influence on actual share ratio.

^b To be read as: The influence of sellers' vertical involvement on actual share ratio will be stronger under conditions of high dynamism than under conditions of low dynamism.

^c To be read as: The influence of richness on actual share ratio will be weaker under conditions of high dynamism than under conditions of low dynamism.

Figures in parentheses are hypothesis numbers.

TABLE III-2**Hypothesized Effects - Dependent Variable Predicted Share Ratio**

Explanatory Variable	Main Effect	Moderating Effects			
		Dynamism	Diversity	Importance	Type Age
Sellers' vertical involvement	+ ^a (2A)	> ^b (11A)	> (14A)	> (17A)	> (20A)
Sellers' lateral involvement	+ (2B)	> (11B)	> (14B)	> (17B)	> (20B)
Buyers' vertical involvement	+ (2C)	> (11C)	> (14C)	> (17C)	> (20C)
Buyers' lateral involvement	+ (2D)	> (11D)	> (14D)	> (17D)	> (20D)
Disclosing	+ (5A)	> (11E)	> (14E)	> (17E)	> (20E)
Initiating	+ (5B)	> (11F)	> (14F)	> (17F)	> (20F)
Signaling	+ (5C)	> (11G)	> (14G)	> (17G)	> (20G)
Frequency	+ (8A)	> (11H)	> (14H)	> (17H)	> (20H)
Richness	+ (8B)	< ^c (11I)	> (14I)	> (17I)	> (20I)

^a To be read as: Sellers' vertical involvement will have a positive influence on predicted share ratio.

^b To be read as: The influence of sellers' vertical involvement on predicted share ratio will be stronger under conditions of high dynamism than under conditions of low dynamism.

^c To be read as: The influence of richness on predicted share ratio will be weaker under conditions of high dynamism than under conditions of low dynamism.

Figures in parentheses are hypothesis numbers.

TABLE III-3

Hypothesized Effects - Dependent Variable Trust

Explanatory Variable	Main Effect	Moderating Effects				
		Dynamism	Diversity	Importance	Type	Age
Sellers' vertical involvement	+ ^a (3A)			> ^b (18A)	> (21A)	< ^c (22A)
Sellers' lateral involvement	+ (3B)			> (18B)	> (21B)	< (22B)
Buyers' vertical involvement	+ (3C)			> (18C)	> (21C)	< (22C)
Buyers' lateral involvement	+ (3D)			> (18D)	> (21D)	< (22D)
Disclosing	+ (6A)	> (12A)	> (15A)	> (18E)	> (21E)	< (22E)
Initiating	+ (6B)	> (12B)	> (15B)	> (18F)	> (21F)	< (22F)
Signaling	+ (6C)	> (12C)	> (15C)	> (18G)	> (21G)	< (22G)
Frequency	+ (9A)			> (18H)	> (21H)	< (22H)
Richness	+ (9B)			> (18I)	> (21I)	< (22I)

^a To be read as: Sellers' vertical involvement will have a positive influence on trust.

^b To be read as: The influence of sellers' vertical involvement on trust will be stronger under conditions of high importance than under conditions of low importance.

^c To be read as: The influence of sellers' vertical involvement on trust will be weaker when age of the relationship is high than when age of the relationship is low.

Figures in parentheses are hypothesis numbers.

IV. METHOD

A. Measures

Descriptions of the measures used in the study appear in Appendix A. Scales used to measure these constructs were either adapted from previous measures or developed specifically for this study. In developing these measures a thorough literature review was first conducted. Next, a pool of items was developed taking into consideration extant measures, the domain of the construct as defined herein, and the context of the study. Preliminary questionnaires were then developed and pre-tested via personal interviews with purchasing managers and purchasing agents. Successive rounds of two or three interviews were conducted, each round incorporating refinements to the instrument based on prior rounds. This process was continued until feedback from the interviewees indicated that the instrument was clear, answerable, meaningful, and relevant, and that further continuation of the process was unlikely to result in substantive improvements. The process comprised four rounds and a total of ten interviews (averaging about 45 minutes each).

In the early stages of questionnaire development, three changes were made to the original plans regarding study variables. First, the emphasis of the variable disclosing was shifted toward sellers' tendencies to reveal potentially

damaging or embarrassing information, rather than straightforward information regarding plans, objectives, and so on (as originally planned). This latter information was viewed by interviewees as "normal-course-of-business," and thus, expected. In contrast, disclosing "less-than-glamorous" information emerged as a more powerful construct in the context of relationships. Second, informants were asked to evaluate a target product (rather than product or service, as originally planned). Early interviews revealed that (1) a "product" focus allowed informants to more easily select a "target" offering, and (2) attempts to elicit a mix of target products and services would be likely to confuse informants. Informants were, however, asked to classify the target product as (1) capital goods, (2) parts or components, (3) raw materials, (4) operating or maintenance supplies, or (5) other. We then used this information to infer the size of the "service" component of the target product. Specifically, we regarded "capital equipment" and "parts and components" as "high service" products, whereas "raw materials" and "operating or maintenance supplies" were classed as "low service." This classification served as the basis for testing moderating effects of product type. Third, an additional dependent measure, satisfaction with the relationship, was incorporated. The purpose of this addition was to assess buyers' affect at a very global level. Accordingly, our measure tapped buyers' overall feelings toward suppliers (including their wishes that other suppliers were "like this one,"

and their general disposition toward continued relationships; see Appendix A). We hypothesized that our global satisfaction measure would be influenced in the same manner as trust, but sought a more global indicator of affect toward the supplier. Analyses and discussion which follow reflect these three changes to prior plans.

B. Data Collection

Data were collected via a national mail survey. The sampling frame for the study was the master membership list of the National Association of Purchasing Management (NAPM), from which a random sample of 800 names and business addresses was obtained (NAPM provided this sample). NAPM members typically belong to their firm's purchasing department and have the title of "Purchasing Manager" or "Purchasing Agent." Along with the study questionnaire, each of the 800 informants was sent a personalized cover letter and postage-paid business reply envelope. No gift or monetary incentive was provided. One week after the initial mailing a postcard reminder was sent to every informant, again requesting their participation in the study. Three weeks after the initial mailing a replacement questionnaire, modified cover letter, and return envelope were mailed to those informants who had not yet responded. Copies of the first cover letter, postcard, and second cover letter appear in

Appendix B. Appendix C contains the study questionnaire.

Thirty-five of the 800 questionnaires were returned unanswered for one of the following reasons: (1) the address was undeliverable, (2) the addressee was no longer employed at that address, or (3) the addressee was no longer employed in a purchasing-related position. Consequently, these informants were deleted from the sample. From the remaining 765 informants, a total of 454 usable responses was obtained for a response rate of 59.3%, which compares quite favorably to response rates typically obtained in organizational surveys. Further, a comparison of the early and late responses (specifically, the first 20% received versus the last 20% received) showed no significant differences in the means of any of the study variables, indicating that nonresponse bias is unlikely to be a problem.

Informants were asked to select their firms' most recent product purchases meeting the following three conditions: (1) the product was at least moderately important, (2) they purchased the product from two or more competing suppliers, and (3) they were very familiar with their firms' relationships with these suppliers. These instructions were intended to ensure that informants had adequate knowledge of the relationship and that the relationship was of sufficient importance that the effects hypothesized in our model could reasonably be expected. Informants were instructed to select one supplier of the product,

thereafter directing their responses to the "target product" and "target supplier." So as to obtain variance on the dependent measures, half of the informants were asked to select as the target supplier the one with the largest share of their business for the target product, and the other half were asked to select the supplier with the smallest share.

C. Data Analysis

First, the reliability of each of the measures was estimated by computing its Cronbach alpha. One scale item with a low inter-item correlation was eliminated (importance scale). Cronbach alphas for the scales range from .67 to .92, generally demonstrating high reliability (see Table IV-1).

Next, a regression model was estimated for each of the dependent measures as follows:

$$\begin{aligned} \text{SHRA} = & b_0 + b_1\text{RATE} + b_2\text{VRTS} + b_3\text{LATS} + b_4\text{VRTB} + b_5\text{LATB} \\ & + b_6\text{DISC} + b_7\text{INIT} + b_8\text{SIG} + b_9\text{FREQ} + b_{10}\text{RICH} \end{aligned} \quad (4.1)$$

$$\begin{aligned} \text{SHRP} = & b_0 + b_1\text{RATE} + b_2\text{VRTS} + b_3\text{LATS} + b_4\text{VRTB} + b_5\text{LATB} \\ & + b_6\text{DISC} + b_7\text{INIT} + b_8\text{SIG} + b_9\text{FREQ} + b_{10}\text{RICH} \end{aligned} \quad (4.2)$$

$$\begin{aligned} \text{TRST} = & b_0 + b_1\text{RATE} + b_2\text{VRTS} + b_3\text{LATS} + b_4\text{VRTB} + b_5\text{LATB} \\ & + b_6\text{DISC} + b_7\text{INIT} + b_8\text{SIG} + b_9\text{FREQ} + b_{10}\text{RICH} \end{aligned} \quad (4.3)$$

$$\begin{aligned} \text{SAT} = & b_0 + b_1\text{RATE} + b_2\text{VRTS} + b_3\text{LATS} + b_4\text{VRTB} + b_5\text{LATB} \\ & + b_6\text{DISC} + b_7\text{INIT} + b_8\text{SIG} + b_9\text{FREQ} + b_{10}\text{RICH} \end{aligned} \quad (4.4)$$

variable names are as follows:

SHRA	- Actual share ratio
SHRP	- Predicted share ratio
TRST	- Trust
SAT	- Satisfaction
RATE	- Product rating
VRTS	- Sellers' vertical involvement
LATS	- Sellers' lateral involvement
VRTB	- Buyers' vertical involvement
LATB	- Buyers' lateral involvement
DISC	- Disclosing
INIT	- Initiating
SIG	- Signaling
FREQ	- Frequency
RICH	- Richness

Parameter estimates for the above regression models provided bases for testing hypothesized main effects. The following summary indicates hypothesis numbers (in parentheses) immediately following their test parameters (refer also to Tables III-1, III-2, and III-3 for a summary of hypothesized main effects):

Model 4.1 (Actual share ratio) -

b_2 (1A), b_3 (1B), b_4 (1C), b_5 (1D), b_6 (4A), b_7 (4B), b_8 (4C), b_9 (7A), b_{10} (7B).

Model 4.2 (Predicted share ratio) -

b_2 (2A), b_3 (2B), b_4 (2C), b_5 (2D), b_6 (5A), b_7 (5B), b_8 (5C), b_9 (8A), b_{10} (8B).

Model 4.3 (Trust) -

b_2 (3A), b_3 (3B), b_4 (3C), b_5 (3D), b_6 (6A), b_7 (6B), b_8 (6C), b_9 (9A), b_{10} (9B).

Model 4.4 (Satisfaction) -

Same effects as model 4.3, hypotheses not formally specified.

Initially, sellers' dependence was to be included as a covariate in the regression model for trust. However, after a preliminary run showed sellers' dependence to have a non-significant coefficient ($p < .93$), and further considering the low (.13) correlation between sellers' dependence and trust, it was dropped from further analysis. It does not appear that sellers' dependence, as measured by the proportion of business that sellers derive from the buyer, is significantly related to buyers' levels of trust.

The presence of outlying observations was investigated following the initial regression runs. Observations which were both outliers (student residuals > 3.0) and influential, as measured by Cook's D, were eliminated (Neter, Wasserman, and Kutner 1983). This procedure resulted in the elimination of two observations from both share ratio (actual and predicted) regressions (in each instance

the same two observations). Subsequently, these regression models were rerun, as well as regressions on three random samples of 80% of the observations. These runs, together with pairwise correlations among the variables (Table IV-2), were used to assess the stability of the parameter estimates.

To assess moderating effects, subgroup analyses were performed (Arnold 1982; Sharma, Durand and Gur-Arie 1981). Observations were first ordered from low to high on a moderator variable (see Table IV-3 for pairwise correlations among the moderator variables). Next, two subgroups were formed corresponding to (1) the low 40% of the observations and (2) the high 40% of the observations (the middle 20% of the observations were deleted so as to obtain greater contrast between the subgroups). Following this, for each model (4.1-4.4 above) in which the moderator was hypothesized to have an effect, separate regressions were run on the low and high subgroups. Again, 80% random samples were estimated for each subgroup. The above subgrouping procedure was repeated for each of the moderator variables. A Chow test (Chow 1960) was performed to test the significance of differences between corresponding regression coefficients in the low and high groups (refer to Appendix D for an explanation of this test). The following example illustrates this procedure for the moderator variable dynamism and the dependent variable actual share ratio:

$$\begin{aligned}
SHRA_{\text{high dynamism}} = & b_{0hd} + b_{1hd}RATE + b_{2hd}VRTS + b_{3hd}LATS + b_{4hd}VRTB \\
& + b_{5hd}LATB + b_{6hd}DISC + b_{7hd}INIT + b_{8hd}SIG \\
& + b_{9hd}FREQ + b_{10hd}RICH
\end{aligned}$$

$$\begin{aligned}
SHRA_{\text{low dynamism}} = & b_{0ld} + b_{1ld}RATE + b_{2ld}VRTS + b_{3ld}LATS + b_{4ld}VRTB \\
& + b_{5ld}LATB + b_{6ld}DISC + b_{7ld}INIT + b_{8ld}SIG \\
& + b_{9ld}FREQ + b_{10ld}RICH
\end{aligned}$$

To test the moderating effects of dynamism, corresponding regression coefficients for the high dynamism and low dynamism subgroups were compared as follows (refer to Tables III-1, III-2, and III-3 for a summary of hypothesized moderating effects, and to Appendix D for an explanation of the Chow test used to compare regression coefficients):

Regression Coefficients (High dynamism - Low dynamism)		Hypothesis
$b_{2hd} - b_{2ld}$		10A
$b_{3hd} - b_{3ld}$		10B
$b_{4hd} - b_{4ld}$		10C
$b_{5hd} - b_{5ld}$		10D
$b_{6hd} - b_{6ld}$		10E
$b_{7hd} - b_{7ld}$		10F
$b_{8hd} - b_{8ld}$		10G
$b_{9hd} - b_{9ld}$		10H
$b_{10hd} - b_{10ld}$		10I

A similar procedure was used for the balance of the moderating effects

involving actual share ratio, and all moderating effects involving predicted share ratio, trust, and satisfaction.

Preliminary analyses of the above estimations revealed two broad patterns. First, none of the parameter coefficients in regressions for share ratio (actual and predicted), including the subgroups, were significant. Adjusted R^2 for regressions for actual share ratio and predicted share ratio averaged .029 and .034 respectively. Second, none of the coefficients for sellers' vertical involvement, sellers' lateral involvement, buyers' vertical involvement, and buyers' lateral involvement (i.e., the boundary structure variables, see Figure III-1) were significant in regressions for trust and satisfaction, including the subgroups. Pairwise correlations in the range of .06 to .13 between these predictors and trust and satisfaction (see Table IV-2) confirm the weak relationship. So as to reduce the standard errors of the other predictor variables, the regression models for trust and satisfaction were rerun, omitting the four boundary structure variables. R^2 (unadjusted) for these models (pooled sample) declined from .6236 to .6196, and .6422 to .6370, respectively. Similar results obtained for the subgroupings. Since the boundary structure variables had essentially no ability to explain additional variance in trust and satisfaction, and were non-significant in all regressions, they were omitted from the final regression models. Accordingly, the final regression models for trust and satisfaction were as follows:

$$\text{TRST} = b_0 + b_1\text{RATE} + b_2\text{DISC} + b_3\text{INIT} + b_4\text{SIG} + b_5\text{FREQ} + b_6\text{RICH} \quad (4.5)$$

$$\text{SAT} = b_0 + b_1\text{RATE} + b_2\text{DISC} + b_3\text{INIT} + b_4\text{SIG} + b_5\text{FREQ} + b_6\text{RICH} \quad (4.6)$$

All analyses involving trust and satisfaction were repeated, in accordance with the procedures described above, using regression models 4.5 and 4.6. The results of these final analyses are reported in Tables IV-4 through IV-12. As Tables IV-5 and IV-10 indicate, the pooled sample parameter estimates in regressions for both trust and satisfaction are quite stable. This, along with the moderate size of pairwise correlations among the independent variables (Table IV-2), suggest that collinearity is not a problem. For the dependent variable trust, age of the relationship was found to moderate the influence of communication frequency, environmental diversity was found to moderate the influence of initiation and product rating, and product importance moderated the influence of product rating (see Table IV-4). For the dependent variable satisfaction, age of the relationship moderated the influence of communication frequency, and environmental diversity moderated the influence of product rating (see Table IV-9). No other moderating effects were found. As Tables IV-6, IV-7, IV-8, IV-11, and IV-12 indicate, parameter estimates are stable for all subgroups in which moderating effects were found. Again, considering the stability of these parameter estimates along with the moderate correlation among the independent

variables, it does not appear that collinearity is a problem.

TABLE IV-1
Cronbach's Alpha

	Initial Scale	# Items	Final Scale	# Items
Trust	.81	7	.81	7
Satisfaction	.92	6	.92	6
Disclosing	.71	6	.71	6
Initiating	.91	6	.91	6
Signaling	.87	6	.87	6
Importance	.67	4	.73	3
Dynamism (frequency) ^a	.83	4	.83	4
Dynamism (effect) ^b	.85	4	.85	4
Diversity (consumer) ^c	.75	4	.75	4
Diversity (business) ^d	.82	4	.82	4
Product Rating	.67	3	.67	3

^a Frequency component of environmental dynamism.

^b Effect component of environmental dynamism.

^c Diversity in consumer markets.

^d Diversity in business markets.

Note: Refer to Appendix A for an explanation of measures.

TABLE IV-2

Correlations - Dependent and Independent Variables

	Actual Share Ratio	Predict Share Ratio	Trust	Satis- faction	Product Rating	Lateral Involve. (Buyer)	Vertical Involve. (Buyer)
Actual share ratio	-						
Predicted share ratio	.87	-					
Trust	.12	.16	-				
Satisfaction	.15	.19	.83	-			
Product rating	.05	.08	.55	.60	-		
Lateral involvement (B)	.06	.03	.06	.06	.01	-	
Vertical involvement (B)	.10	.07	.13	.11	.07	.63	-
Lateral involvement (S)	.15	.13	.13	.08	.03	.67	.59
Vertical involvement (S)	.15	.11	.12	.09	.04	.55	.66
Disclosing	.07	.07	.55	.53	.35	.03	.02
Initiating	.18	.19	.66	.64	.41	.16	.21
Signaling	.10	.10	.64	.65	.51	.07	.11
Frequency	.11	.11	.32	.35	.12	.30	.30
Richness	.06	.05	.07	.09	.12	.03	.00
	Lateral Involve. (Seller)	Vertical Involve. (Seller)	Disclos- ing	Initiat- ing	Signal- ing	Freq- uency	Rich- ness
Lateral involvement (S)	-						
Vertical involvement (S)	.70	-					
Disclosing	.06	.08	-				
Initiating	.20	.24	.42	-			
Signaling	.13	.17	.52	.56	-		
Frequency	.26	.34	.14	.47	.28	-	
Richness	-.05	.07	.03	.15	.10	.20	-

TABLE IV-3
Correlations - Moderator Variables

	Age	Importance	Dynamism	Diversity
Age	--			
Importance	.12	--		
Dynamism	.00	.19	--	
Diversity	.11	.06	.12	--

TABLE IV-4
Dependent Variable Trust
Unstandardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Full (N = 435)	.62	.58 ^a	.23 ^a	.32 ^a	.20 ^a	.11 ^d	-2.53 ^d
Low Age ¹ (N = 176)	.66	.57 ^a	.16 ^c	.23 ^a	.31 ^a	.33 ^a	-5.22 ^c
High Age (N = 172)	.61	.63 ^a	.21 ^a	.39 ^a	.18 ^a	-.09 ^d	-.41 ^d
Low Diversity ² (N = 154)	.64	.36 ^a	.21 ^a	.38 ^a	.20 ^a	.07 ^d	-3.47 ^d
High Diversity (N = 158)	.70	.83 ^a	.25 ^a	.19 ^a	.27 ^a	.22 ^c	-4.59 ^d

¹ Coefficient of Frequency greater in low age group (Chow test: $F = 8.93^b$).

² Coefficient of Initiating greater in low diversity group (Chow test: $F = 4.80^c$).
Coefficient of Product Rating greater in high diversity group (Chow test: $F = 5.98^c$).

^a $p < .001$

^b $p < .01$

^c $p < .05$

^d not significant at .05 level

TABLE IV-4 (continued)
Dependent Variable Trust
Unstandardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Low Dynam. (N = 162)	.69	.61 ^a	.25 ^a	.37 ^a	.20 ^b	.07 ^d	-3.46 ^d
High Dynam. (N = 167)	.59	.63 ^a	.23 ^a	.25 ^a	.27 ^a	.06 ^d	-5.36 ^c
Low Import. ³ (N = 174)	.68	.39 ^b	.29 ^a	.43 ^a	.17 ^b	.13 ^d	-2.15 ^d
High Import. (N = 174)	.62	.77 ^a	.13 ^b	.29 ^a	.19 ^b	.09 ^d	-2.16 ^d
Low Service (N = 226)	.64	.72 ^a	.24 ^a	.30 ^a	.12 ^d	.21 ^c	-1.66 ^d
High Service (N = 175)	.61	.55 ^a	.23 ^a	.35 ^a	.24 ^a	-.01 ^d	-4.15 ^d

³ Coefficient of Product Rating greater in high importance group (Chow test: $F = 4.01^c$).

^a $p < .001$

^b $p < .01$

^c $p < .05$

^d not significant at .05 level

TABLE IV-5
Dependent Variable Trust - Pooled Sample
Standardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Full (N = 433)	.62	.24 ^a	.19 ^a	.35 ^a	.20 ^a	.06 ^d	-.05 ^d
80% (N = 347)	.59	.26 ^a	.17 ^a	.35 ^a	.19 ^a	.05 ^d	-.06 ^d
80% (N = 347)	.61	.26 ^a	.21 ^a	.35 ^a	.17 ^a	.05 ^d	-.04 ^d
80% (N = 346)	.63	.23 ^a	.21 ^a	.35 ^a	.21 ^a	.03 ^d	-.01 ^d

^a p<.001

^b p<.01

^c p<.05

^d not significant at .05 level

TABLE IV-6
Dependent Variable Trust - Subgroupings by Age
Standardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Low Age (N = 176)	.66	.24 ^a	.12 ^c	.26 ^a	.31 ^a	.18 ^a	-.11 ^c
80% (N = 142)	.69	.22 ^a	.18 ^b	.25 ^a	.33 ^b	.16 ^a	-.08 ^d
80% (N = 141)	.65	.22 ^a	.26 ^a	.30 ^a	.30 ^a	.19 ^b	-.10 ^d
80% (N = 140)	.67	.23 ^a	.12 ^c	.25 ^a	.30 ^a	.21 ^a	-.11 ^c
High Age (N = 172)	.62	.27 ^a	.18 ^b	.41 ^a	.17 ^b	-.05 ^d	-.01 ^d
80% (N = 135)	.63	.27 ^a	.18 ^b	.44 ^a	.15 ^c	-.04 ^d	.00 ^d
80% (N = 138)	.64	.28 ^a	.20 ^b	.41 ^a	.18 ^c	-.07 ^d	.01 ^d
80% (N = 140)	.63	.30 ^a	.18 ^b	.40 ^a	.17 ^c	-.04 ^d	-.04 ^d

^a p < .001

^b p < .01

^c p < .05

^d not significant at .05 level

TABLE IV-7
Dependent Variable Trust - Subgroupings by Diversity
Standardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Low Diversity (N = 154)	.64	.17 ^b	.19 ^b	.42 ^a	.20 ^b	.04 ^d	-.07 ^d
80% (N = 123)	.63	.16 ^c	.19 ^c	.42 ^a	.23 ^b	-.03 ^d	-.12 ^c
80% (N = 123)	.68	.16 ^c	.25 ^a	.39 ^a	.17 ^c	.11 ^d	-.05 ^d
80% (N = 122)	.65	.15 ^c	.17 ^c	.41 ^a	.25 ^b	.03 ^d	-.09 ^d
High Diversity (N = 158)	.70	.35 ^a	.20 ^a	.22 ^a	.27 ^a	.11 ^c	-.09 ^d
80% (N = 128)	.68	.32 ^a	.24 ^a	.20 ^b	.25 ^b	.09 ^d	-.07 ^d
80% (N = 125)	.72	.30 ^a	.20 ^a	.21 ^b	.32 ^a	.13 ^c	-.11 ^c
80% (N = 127)	.73	.36 ^a	.15 ^b	.28 ^a	.27 ^a	.10 ^d	-.09 ^d

^a p < .001

^b p < .01

^c p < .05

^d not significant at .05 level

TABLE IV-8
Dependent Variable Trust - Subgroupings by Product Importance
Standardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Low Import. (N = 174)	.68	.16 ^b	.24 ^a	.45 ^a	.17 ^b	.07 ^d	-.04 ^d
80% (N = 140)	.69	.16 ^b	.28 ^a	.42 ^a	.19 ^b	.02 ^d	-.05 ^d
80% (N = 140)	.69	.20 ^a	.20 ^a	.43 ^a	.15 ^c	.14 ^c	-.04 ^d
80% (N = 140)	.65	.12 ^c	.22 ^a	.49 ^a	.15 ^c	.06 ^d	-.05 ^d
High Import. (N = 174)	.62	.33 ^a	.12 ^c	.35 ^a	.19 ^b	.05 ^d	-.05 ^d
80% (N = 140)	.66	.36 ^a	.11 ^d	.36 ^a	.19 ^b	.04 ^d	-.09 ^d
80% (N = 138)	.59	.34 ^a	.07 ^d	.34 ^a	.19 ^b	.02 ^d	.02 ^d
80% (N = 138)	.57	.29 ^a	.17 ^c	.31 ^a	.18 ^c	.09 ^d	-.04 ^d

^a p < .001

^b p < .01

^c p < .05

^d not significant at .05 level

TABLE IV-9
Dependent Variable Satisfaction
Unstandardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Full (N = 435)	.64	.80 ^a	.22 ^a	.26 ^a	.20 ^a	.21 ^a	-1.74 ^d
Low Age ¹ (N = 176)	.68	.72 ^a	.24 ^b	.24 ^a	.28 ^a	.42 ^a	-4.96 ^c
High Age (N = 172)	.60	.87 ^a	.27 ^a	.25 ^a	.11 ^d	-.04 ^d	1.18 ^d
Low Diversity ² (N = 154)	.65	.53 ^a	.13 ^d	.30 ^a	.29 ^a	.29 ^b	-1.24 ^d
High Diversity (N = 158)	.68	.93 ^a	.24 ^a	.20 ^a	.21 ^b	.20 ^d	-4.17 ^d

¹ Coefficient of Frequency greater in low age group (Chow test: $F = 10.63^b$).

² Coefficient of Product Rating greater in high diversity group (Chow test: $F = 4.02^c$).

^a $p < .001$

^b $p < .01$

^c $p < .05$

^d not significant at .05 level

TABLE IV-9 (continued)
Dependent Variable Satisfaction
Unstandardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Low Dynam. (N = 162)	.70	.78 ^a	.21 ^a	.33 ^a	.18 ^b	.18 ^d	-.53 ^d
High Dynam.	.60	.95 ^a	.25 ^a	.18 ^b	.18 ^c	.20 ^d	-3.70 ^d
Low Import. (N = 174)	.65	.60 ^a	.24 ^b	.36 ^a	.18 ^b	.29 ^b	-1.31 ^d
High Import. (N = 174)	.63	.98 ^a	.25 ^a	.22 ^a	.12 ^d	.12 ^d	.66 ^d
Low Service (N = 226)	.60	.92 ^a	.22 ^a	.23 ^a	.15 ^c	.24 ^c	-2.82 ^d
High Service (N = 175)	.69	.77 ^a	.23 ^a	.31 ^a	.22 ^a	.18 ^d	-.92 ^d

^a p < .001

^b p < .01

^c p < .05

^d not significant at .05 level

TABLE IV-10
Dependent Variable Satisfaction - Pooled Sample
Standardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Full (N = 433)	.64	.33 ^a	.18 ^a	.28 ^a	.19 ^a	.11 ^a	-.03 ^d
80% (N = 347)	.62	.33 ^a	.17 ^a	.29 ^a	.18 ^a	.11 ^b	-.04 ^d
80% (N = 347)	.63	.34 ^a	.17 ^a	.27 ^a	.19 ^a	.12 ^b	-.01 ^d
80% (N = 346)	.64	.33 ^a	.19 ^a	.26 ^a	.18 ^a	.10 ^b	-.01 ^d

^a p < .001

^b p < .01

^c p < .05

^d not significant at .05 level

TABLE IV-11
Dependent Variable Satisfaction - Subgroupings by Age
Standardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Low Age (N = 176)	.68	.28 ^a	.17 ^b	.25 ^a	.25 ^a	.21 ^a	-.09 ^c
80% (N = 142)	.73	.30 ^a	.18 ^b	.25 ^a	.24 ^a	.25 ^a	-.11 ^c
80% (N = 141)	.69	.27 ^a	.20 ^b	.23 ^a	.25 ^a	.21 ^a	-.08 ^d
80% (N = 140)	.72	.24 ^a	.15 ^c	.29 ^a	.28 ^a	.20 ^a	-.09 ^d
High Age (N = 172)	.60	.39 ^a	.25 ^a	.27 ^a	.11 ^d	-.02 ^d	.02 ^d
80% (N = 135)	.59	.37 ^a	.24 ^a	.29 ^a	.12 ^d	.02 ^d	-.01 ^d
80% (N = 138)	.62	.38 ^a	.27 ^a	.29 ^a	.08 ^d	-.01 ^d	.01 ^d
80% (N = 140)	.66	.40 ^a	.27 ^a	.28 ^a	.13 ^c	-.07 ^d	.03 ^d

^a p < .001

^b p < .01

^c p < .05

^d not significant at .05 level

TABLE IV-12
Dependent Variable Satisfaction - Subgroupings by Diversity
Standardized Parameter Estimates

Sample	R ²	Product Rating	Disclosing	Initiating	Signaling	Frequency	Richness
Low Diversity (N = 154)	.65	.23 ^a	.11 ^d	.31 ^a	.27 ^a	.15 ^b	-.02 ^d
80% (N = 123)	.68	.24 ^a	.11 ^d	.34 ^a	.30 ^a	.10 ^d	-.07 ^d
80% (N = 123)	.68	.19 ^b	.18 ^b	.31 ^a	.23 ^b	.19 ^b	.01 ^d
80% (N = 122)	.64	.22 ^b	.09 ^d	.34 ^a	.27 ^a	.14 ^c	-.03 ^d
High Diversity (N = 158)	.68	.39 ^a	.19 ^a	.23 ^a	.21 ^b	.10 ^d	-.08 ^d
80% (N = 128)	.67	.39 ^a	.19 ^b	.28 ^a	.17 ^c	.04 ^d	-.09 ^d
80% (N = 125)	.68	.39 ^a	.20 ^b	.22 ^b	.20 ^b	.14 ^c	-.06 ^d
80% (N = 127)	.68	.40 ^a	.17 ^b	.25 ^a	.19 ^b	.11 ^d	-.09 ^d

^a p < .001

^b p < .01

^c p < .05

^d not significant at .05 level

V. FINDINGS AND DISCUSSION

A. Main Effects

1. Actual and Predicted Share Ratio. On balance, our findings suggest that boundary behavior has little ability to predict sellers' shares. Specifically, since none of the regression coefficients for model 4.1 were significant, hypotheses 1A, 1B, 1C, 1D, 4A, 4B, 4C, 7A, and 7B, relating to the main effects on actual share ratio, were not supported. Likewise, none of the regression coefficients for model 4.2 were significant, thus failing to support hypotheses 2A, 2B, 2C, 2D, 5A, 5B, 5C, 8A, and 8B, relating to the main effects on predicted share ratio. In sharp contrast to these findings, it appears that boundary behavior is quite effective in explaining buyers' trust and satisfaction (discussed below). In light of these latter findings, the former are surprising because trust in, and satisfaction with, a seller might, on their own merits, reasonably be expected to lead to rewarding that seller with greater business. Parenthetically, we tested this supposition by regressing both actual share ratio and predicted share ratio on trust and satisfaction, and neither variable was significant in explaining sellers' shares (actual or predicted). It is also important to bear in mind that product rating was non-significant in regressions for sellers' shares (both actual and predicted).

These results suggest that other factors, not addressed in the study, underlie buyers' failure to "reward" those suppliers which they trust, and with which they are satisfied, and whose product they rate highly. Clearly, buyers may choose to reward suppliers in ways other than increasing shares (e.g., paying a premium price, "trading up" to high-margin goods, accepting seller-preferred delivery/service schedules, and so on). Also, non-performance factors may influence sellers' shares (e.g., sellers' location, complementary offerings, reciprocity, and so on). On the other hand, it may be that control mechanisms necessary to effect "performance-share" matching are not in place in many organizations, or that momentum and convenience tend to perpetuate buying patterns even though disconfirming evidence is available. The concluding section of the dissertation discusses these issues more fully.

Notwithstanding the foregoing discussion, which addresses omitted variables which might be responsible for the "non-findings" relating to sellers' shares, methodological limitations must also be considered. It should be noted that the study employed a single informant in each organization. Consequently, it is not possible to fully assess the construct validity of measures employed in the study, particularly convergent validity (Campbell and Fiske 1959; Phillips 1981). Had multiple methods (multiple informants) been utilized for each organizational measure, such an assessment would be possible. This limitation was knowingly

accepted in the study because the chosen organizational informants were those individuals thought to have greatest access to, and knowledge of, the requested information. Nonetheless, this is an important limitation.

2. Trust and Satisfaction. Because the regression coefficients of sellers' vertical involvement, sellers' lateral involvement, buyers' vertical involvement, and buyers' lateral involvement in model 4.3 (original trust model) were all non-significant, hypotheses 3A, 3B, 3C, and 3D (Table III-1) were not supported. In regression model 4.5 (final trust model), coefficients of disclosing, initiating, and signaling were each significant, and in the hypothesized direction (see Table IV-5), thus supporting hypotheses 6A, 6B, and 6C respectively. Coefficients for frequency and richness were not significant; hypotheses 9A and 9B were, therefore, not supported. As expected, product rating also had a strong positive relationship to trust (Table V-1 summarizes findings resulting from analysis of the final trust model). Overall, then, our findings suggest that product rating, disclosing, initiating, and signaling have a positive, and important, influence on buyers' trust (R^2 for the final trust model = .62, see Table IV-4). Initiation appears to have the greatest influence, moderately greater than product rating, disclosing, and signaling, which are all of roughly equal influence (see full sample standardized parameter estimates, Table IV-5). In general, sellers who reveal

the less-than-glamorous aspects of their businesses, make overt efforts to keep abreast of their buyers' changing businesses, and clearly communicate their intentions, appear to be trusted to a greater degree than sellers who do not do these things as well or to as great a degree. These findings should be of interest to both researchers and managers. We extend prior research findings in support of a communication --> trust linkage (Anderson and Narus 1990; Anderson and Weitz 1989; Loomis 1959), by identifying specific kinds of communication behavior which appear to be influential. For managers, our findings provide actionable bases for "effective" communication.

Regression coefficients of sellers' vertical involvement, sellers' lateral involvement, buyers' vertical involvement, and buyers' lateral involvement in model 4.4 (original satisfaction model) were all non-significant. Accordingly, our prior expectation of a positive relationship between these variables and satisfaction was not supported. In regression model 4.6 (final satisfaction model) coefficients of product rating, disclosing, initiating, signaling, and frequency are all significant, and positive, in support of our prior expectations. Finally, the coefficient of richness is non-significant, failing to support our expectation (Table V-2 summarizes findings resulting from analysis of the final satisfaction model). These findings suggest that, overall, satisfaction with the relationship is positively, and substantially (R^2 for the final satisfaction model = .64, see Table IV-9),

influenced by product rating, disclosing, initiating, signaling, and communication frequency. Product rating and initiating appear to have the greatest influence on satisfaction, followed by disclosing and signaling, followed by frequency (see full sample standardized parameter estimates, Table IV-10). In general, satisfaction with the relationship and trust appear to be influenced similarly, with the exception that communication frequency has a positive overall effect on satisfaction (but not trust). On balance, then, it appears that sellers can employ essentially the same communication strategy to enhance both buyers' trust and their overall satisfaction with the relationship. The single point of difference appears to be that increasing communication frequency may be more effective in enhancing satisfaction.

The above discussion must be qualified, however, by noting the cross-sectional nature of the study. This is an important limitation which significantly tempers our ability to make causal attributions. It is possible, for example, that more frequent communication, as well as greater disclosure, initiating, and signaling tend to occur in those relationships which are characterized by greater trust and satisfaction (i.e., reverse causation). Furthermore, although our *a priori* reasoning was that greater richness would lead to greater trust and satisfaction, it is possible that low trust and satisfaction ("crisis levels") could lead to buyer and seller engaging in greater face-to-face discussions (to "fix" the problem).

Due to the cross-sectional nature of the study, had both processes taken place in our sample, richness would appear to be unrelated to trust (or satisfaction) as we observed. Consequently, our findings, both in support of and contrary to *a priori* reasoning, should be interpreted cautiously.

B. Moderating Effects

1. Actual and Predicted Share Ratio. None of the subgroup regression coefficients in models 4.1 (actual share ratio) and 4.2 (predicted share ratio) were significant for subgroupings on any of the moderator variables; consequently no moderating effects were found. Accordingly hypotheses 10A-I, 13A-I, 16A-I, and 19A-I (Table III-1), regarding moderating effects on actual share ratio, were not supported. Similarly, hypotheses 11A-I, 14A-I, 17A-I, and 20A-I (Table III-2) were not supported. Given the absence of main effects on actual and predicted share ratios, and the low explained variance ($R^2 = .029$ and $.034$, respectively), these results are not surprising. The previously noted limitations regarding omitted variables, and the use of a single informant, might also have contributed to these "non-findings."

2. Trust and Satisfaction. As noted earlier, the subgroup regression coefficients of sellers' vertical involvement, sellers' lateral involvement, buyers'

vertical involvement, and buyers' lateral involvement, in models 4.3 (original trust model) and 4.4 (original satisfaction model) were all non-significant. Accordingly, hypotheses 18A-D, 21A-D, and 22A-D, regarding moderating effects of the influence of these variables on trust (Table III-3), were not supported, nor were our expectations of similar moderating effects regarding satisfaction. Findings regarding the remaining moderating effects involving trust and satisfaction, obtained from subgroup analyses of the final trust model (4.5) and the final satisfaction model (4.6), are discussed below and summarized in Tables V-1 and V-2.

First, concerning environmental dynamism, no moderating effects were found on the relationship between trust and disclosing, initiating, or signaling. Thus, hypotheses 12A, 12B, and 12C, respectively, were not supported. Second, concerning environmental diversity, no moderating effects were found on the relationship between trust and disclosing or signaling; therefore, hypotheses 15A and 15C, respectively, were not supported. Environmental diversity moderates the relationship between trust and initiating, but contrary to our hypothesis 15B. Also, environmental diversity moderates the relationship between trust and product rating, such that the relationship is stronger when environmental diversity is high. Third, concerning product importance, no moderating effects were found on the relationship between trust and disclosing, initiating, signaling,

frequency, or richness; thus, hypotheses 18E, 18F, 18G, 18H, and 18I are not supported. Trust and product rating are, however, more strongly related when product importance is high. Fourth, concerning product type, no moderating effect was found on the relationship between trust and disclosing, initiating, signaling, frequency, or richness; thus, hypotheses 21E, 21F, 21G, 21H, and 21I are not supported. Finally, concerning age of the relationship, frequency is more strongly related to trust when age is low, thus supporting hypothesis 22H. Age does not, however, moderate the relationship between trust and disclosing, initiating, signaling, or richness. Hypotheses 22E, 22F, 22G, and 22I are, therefore, not supported. Refer to Table IV-4 for the results of those Chow tests significant at the .05 level.

Findings regarding moderating effects on satisfaction are quite similar to those for trust. The only two exceptions are as follows: (1) environmental diversity has no moderating effect on the relationship between satisfaction and initiating (whereas, trust and initiating are more strongly related when diversity is low), and (2) product importance has no moderating effect on the relationship between satisfaction and product rating (whereas, trust and product rating are more strongly related when product importance is high). All other moderating effects are similar to trust. Refer to Table IV-9 for the results of those Chow tests significant at the .05 level.

Our findings suggest that when age of the relationship is low, communication frequency has a greater positive influence on trust. Similarly, age of the relationship also appears to moderate the influence of communication frequency on satisfaction. Apparently, communication frequency is more influential in the early stages of a relationship, when initial perceptions are being formed. Later on perceptions may become "calcified" and relatively resistant to modification. Sellers may be particularly interested in this finding because it is easy for them to distinguish among customers according to age. Therefore, it should prove useful for them to design communication programs separately for different age groups, with more "intensive" communication strategies directed to the newer customers.

Our findings also suggest that environmental diversity moderates the influence of both initiating and product rating on buyers' trust. When environmental diversity is low, initiating appears to have a greater influence on trust than when environmental diversity is high. As noted, this finding runs counter to our hypothesis. Since our definition of environmental diversity focuses on the level of diversity in buyers' output markets (customers), it is logical to expect that, in instances of high diversity, initiating behavior on the part of sellers (efforts to keep abreast of buyers' customers and competitive environment) would have a greater influence on buyers' trust. However, the opposite was

found in our sample, i.e., the influence of initiating on trust appears to be weaker when environmental diversity is high. In search of possible explanations for this finding, we first note that the influence of product rating on buyers' trust was also moderated by environmental diversity, but in the opposite direction. That is, product rating was found to have a greater positive influence on buyers' trust for the high diversity group. Collinearity between product rating and initiating could, therefore, lead to the unexpected finding regarding initiating. However, the correlation between product rating and initiating is only .41 (Table IV-2), and this, when considered with the relatively stable subgroup parameter estimates (Table IV-7), suggests that collinearity is unlikely to have been responsible. Another possible explanation for this finding is that buyers' facing diverse markets produce a wider range of products, have more suppliers, purchase a wider variety of inputs, and thus, on average, attach less importance to a single product/supplier than do buyers facing less diverse markets. This possibility was investigated and rejected. No significant difference in the level of target product importance was found between the high diversity and low diversity groups; this is confirmed by the very low correlation (.06) between environmental diversity and target product importance. Finally, buyers in firms facing very diverse markets (serving numerous customers unrelated to the target product via the value chain) may view initiating (attempts by suppliers to "find

out" about their customers and competitive situation) as rather irrelevant. This explanation is, however, highly speculative, and further research would appear to be necessary before drawing any such conclusion.

Our failure to find other moderating effects is not offered as evidence that the other moderator variables are unrelated to boundary behavior. For example, buyers' levels of trust and satisfaction may tend to vary systematically across specific product categories, within our broad classes of product type. Accordingly, the variance due to specific category effects might have overwhelmed boundary behavior effects, thus leading to our finding that product type has no moderating influence. Further, dimensions such as complexity and intangibility may be the most significant product-related factors bearing on boundary behavior. Our four broad classes of product type may not have cleanly tapped differences on these dimensions. A second example concerns environmental dynamism, where, again no moderating effects were observed. It is possible that under conditions of high dynamism, a great deal of boundary behavior "noise" is introduced (that is, excessive, ineffective communication), stimulated by the turbulent environment. The presence of such "noise" would be likely to confound the moderating effect of dynamism on the influence of communication frequency, because the relative efficiency of communication may differ across the high/low dynamism conditions. Clearly, a number of other explanations might

be offered for these, and other, hypothesized moderating effects which we failed to observe. Accordingly, we again note that our findings should be interpreted cautiously.

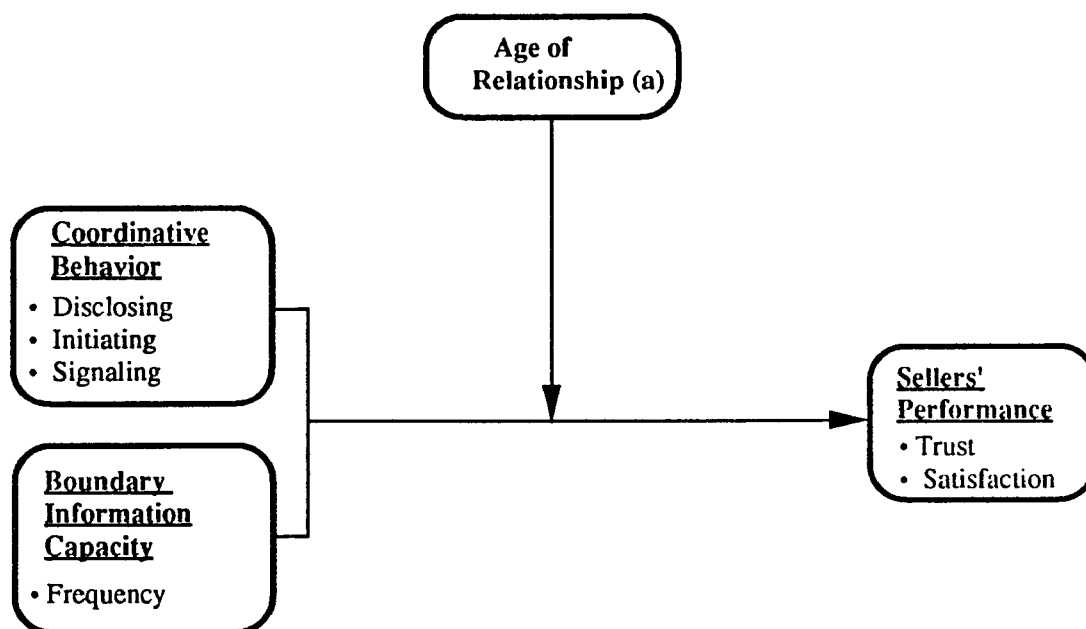
C. Summary

In summary, our findings suggest that, in addition to product rating, the boundary behaviors, (1) disclosing, (2) initiating, and (3) signaling, have, overall, a positive influence on buyers' trust. Furthermore, when the age of the buyer-seller relationship is low, communication frequency has a significant, and significantly greater, positive influence on trust than when age of the relationship is high. Overall, the boundary behaviors, (1) disclosing, (2) initiating, (3) signaling, and (4) frequency, appear to have a positive influence on buyers' satisfaction. Age of the relationship also moderates the influence of communication frequency on buyers' satisfaction; when age is low the influence of communication frequency on satisfaction is stronger than when age is high. Figure V-1 presents a revised model of the influence of boundary behavior on buyers' trust and satisfaction, based on the findings of the study. Our study thus extends prior research by identifying specific communication behaviors which lead to trust and satisfaction. Managers should also be interested because these boundary behaviors represent actionable interaction strategies. This finding

points to the importance of boundary behavior in the early stages of relationship development, and suggests to managers that newer customers could usefully be targeted with more intensive communication strategies.

Finally, it should be noted that environmental diversity appears to moderate the influence of initiating on buyers' trust. When diversity is high, the effect of initiating on trust appears, unexpectedly, to be weaker than when diversity is low. Given the unexpected direction of this effect, and lack of compelling explanation for it, it seems premature to offer recommendations without further investigation.

FIGURE V-1
Revised Model of Sellers' Performance in Business Relationships



(a) Age has a moderating effect on the influence of frequency only.

TABLE V-1
Dependent Variable Trust
Summary of Findings

Explanatory Variable	Main Effect	Moderating Effects				
		Dynamism	Diversity	Importance	Type	Age
Disclosing	+ ^a (6A) ^s	-- (12A) ^{ns}	-- (15A) ^{ns}	-- (18E) ^{ns}	-- (21E) ^{ns}	-- (22E) ^{ns}
Initiating	+ (6B) ^s	-- (12B) ^{ns}	< ^b (15B) ^{ns}	-- (18F) ^{ns}	-- (21F) ^{ns}	-- (22F) ^{ns}
Signaling	+ (6C) ^s	-- (12C) ^{ns}	-- (15C) ^{ns}	-- ^d (18G) ^{ns}	-- (21G) ^{ns}	-- (22G) ^{ns}
Frequency	-- (9A) ^{ns}	--	--	-- (18H) ^{ns}	-- (21H) ^{ns}	< (22H) ^s
Richness	-- (9B) ^{ns}	--	--	-- (18I) ^{ns}	-- (21I) ^{ns}	-- (22I) ^{ns}
Product Rating	+	--	> ^c	>	--	--

^a To be read as: Disclosing was found to have a positive influence on trust.

^b To be read as: The influence of initiating on trust was found to be weaker when diversity was high than when diversity was low.

^c To be read as: The influence of product rating on trust was found to be stronger when product importance was high than when product importance was low.

^d Effect not significant at .05 level.

^s Hypothesis supported.

^{ns} Hypothesis not supported.

TABLE V-2
Dependent Variable Satisfaction
Summary of Findings

Explanatory Variable	Main Effect	Moderating Effects				
		Dynamism	Diversity	Importance	Type	Age
Disclosing	+ ^a	-- ^d	--	--	--	--
Initiating	+	--	--	--	--	--
Signaling	+	--	--	--	--	--
Frequency	+	--	--	--	--	< ^c
Richness	--	--	--	--	--	--
Product Rating	+	--	> ^b	--	--	--

^a To be read as: Disclosing was found to have a positive influence on satisfaction.

^b To be read as: The influence of product rating on satisfaction was found to be stronger when diversity was high than when diversity was low.

^c To be read as: The influence of frequency on satisfaction was found to be weaker when age of the relationship was high than when age of the relationship was low.

^d Effect not significant at .05 level.

VI. CONCLUSION

We have examined how boundary behavior influences buyers' trust in, and satisfaction with, suppliers. We have also examined the influence of boundary behavior on the share of business received by suppliers in competitive situations. Further, we have examined how effects of the environment, buying situation, and age of the relationship moderate these influences.

In sum, boundary behavior is quite effective in explaining variance in buyers' levels of trust and satisfaction, but is ineffective in explaining variance in the amount of business received by suppliers. The age of the buyer-seller relationship moderates the influence of communication frequency on both trust and satisfaction. In each case communication frequency has a stronger (positive) influence when age of the relationship is relatively low, thus pointing to the importance of maintaining an "open" channel of communication in the early stages of relationships. This finding should be important to managers who have "key" account responsibility; age of the relationship may be a variable which is overlooked by many of them and we provide evidence that newer relationships may benefit most from intensive communication strategies. Environmental diversity was found to moderate the influence of initiation, but, surprisingly, in the opposite direction from that which we had predicted. As noted earlier, it

does not appear prudent to draw conclusions from this finding.

Of particular interest are the "non-findings" relating to sellers' shares of business. This area would seem to be a particularly fruitful one for future research. Prior research on market share has concentrated heavily on consumer markets, primarily retail grocery (e.g, Guadagni and Little 1983; Lattin and Bucklin 1989). Recently, this research has benefitted from scanner data which facilitates investigation at the level of the individual consumer. In the sparse literature on competition in business markets, however, no prior studies have investigated sellers' shares, or their antecedents, at the individual account level. It appears that an understanding of buyers' allocation behavior in competitive supplier situations would fill an important void in our knowledge of industrial buyer behavior. Our initial attempt at filling this void suggests several avenues that should be pursued in future research.

First, although we defined sellers' performance in terms of share, follow-up research should examine the notion of "reward," as seen from the buyer's perspective. Rather than attempting to explain share in isolation from other reward factors, it will probably prove more useful to examine the behavior of a set of variables which captures this notion. In addition to share, suggestions advanced earlier include (1) price premium, (2) favoring high-margin items, and (3) accepting seller-preferred delivery/service schedules. Second, reward

structures are likely to vary by industry, therefore, future research should examine them by industry. Third, "non-performance" factors such as sellers' location, complementary offerings, and reciprocity are likely to have an effect on the manner in which buyers reward sellers. These factors were not considered in this study. Therefore, a model predicting sellers' rewards might include both a set of "performance" antecedents (those in this study) and a set of "non-performance" antecedents. Fourth, proceeding in a somewhat different direction, the issue of performance-reward matching (control) is raised by the findings of our study. We did not assess whether buyers in our sample had in-place supplier rating systems. Such systems range from very informal methods to elaborate, computer-based "cost-ratio" plans (Hutt and Speh 1989). It would be interesting to examine whether share, under any of these systems, is differentially predicted by supplier evaluations. To date, no study has addressed this question.

APPENDIX A

Measures

Share Ratio - Actual

1. In all, how many suppliers did you use for the target product during the past year?
2. What percent of last year's target product purchases (dollar value) were from the target supplier?

Computed as the ratio of target supplier's actual share, as reported by buyer, to "expected share" (expected share = $1/n$, where n = number of suppliers used by buyer for target product).

Share Ratio - Predicted

1. Assuming you were not bound by contractual or other constraints (and the target supplier had no supply limitations), what share would the target supplier have received?

Computed as the ratio of target supplier's predicted share to expected share (above).

Trust

1. We are confident they would charge us fair prices even if they had an opportunity to take advantage of us.
2. We're afraid they would cut corners if we let up on inspection.
3. If we were facing unprofitable times, we believe that they would share the burden by trimming their margins.
4. If it was important to us, we could count on them to do a rush job even if they had to absorb some additional costs.
5. They readily acknowledge it if another supplier can meet our needs better.
6. We are confident in their ability to help us solve any problems involving the target product.
7. From our perspective, this supplier is very competent.

Items scored on a 5-point scale ranging from "strongly disagree" to "strongly agree."

Satisfaction with Relationship

1. We are delighted with our overall relationship with them.
2. We wish more of our suppliers were like this one.
3. We would like our relationship with them to continue in the coming years.
4. It is a pleasure dealing with this supplier.
5. There is always some problem or another with this supplier.
6. There are close ties between us and this supplier.

Items scored on a 5-point scale ranging from "strongly disagree" to "strongly agree."

Buyers' Lateral Involvement

Please give 10 points to the department in **your firm** which communicated most frequently with the target supplier over the past year, then give 0 to 9 points to the remaining departments. (An illustrative example followed.)

1. Sales/Marketing
2. Production/Quality Control
3. R&D/Engineering
4. Finance/Accounting
5. Transportation/Distribution
6. Purchasing/Materials Management

Computed as the number of non-zero departments. Adapted from Johnston and Bonoma 1981.

Buyers' Vertical Involvement

Please give 10 points to the level in **your firm** which communicated most frequently with the target supplier over the past year, and 0 to 9 points to the remaining levels, as above.

1. Top Management (Directors, Pres., Exec. VP, and so on)
2. Functional Vice President
3. Department Manager (Sales Mgr., Purch. Mgr., and so on)
4. Department Members (Sales Rep., Purch Agent, and so on)
5. Clerical

Computed as the number of non-zero levels. Adapted from Johnston and Bonoma 1981.

Sellers' Lateral Involvement

Now, please focus on the **target supplier** and give 10 points to the department in their company which communicated most frequently with your firm over the past year, and 0 to 9 points to the remaining departments.

Same scale and computation as for buyers' lateral involvement.

Sellers' Vertical Involvement

Please give 10 points to the level in the **target supplier** which communicated most frequently with your firm over the past year, and 0 to 9 points to the remaining levels.

Same scale and computation as for buyers' vertical involvement.

Disclosing

1. With respect to the target supplier, it can be said that they . . .
 - .. would try to cover it up if they had a manufacturing setback.
 - .. tend to be secretive about politics in their company.
 - .. would freely admit it if they were having internal labor problems.
 - .. tend to sidestep talk about weaknesses in their organization.
 - .. are forthcoming in discussing managerial promotions and changes in their company.
 - .. give us a clear picture of what goes on behind the scenes in their company.

Items scored on a 5-point scale ranging from "strongly disagree" to "strongly agree."

Initiating

1. Keeping in mind your overall relationship with the target supplier, please indicate the **extent** to which they . . .
 - .. spend time trying to learn how they can help you lower the cost of using the target product.
 - .. spend time trying to help you gain maximum benefit from the target product.
 - .. make an effort to keep abreast of your customers' changing needs.
 - .. show an interest in helping you become more competitive.
 - .. keep up with changes in your competitive environment.
 - .. show interest in understanding how your firm works.

Items scored on 5-point scales ranging from "not at all" to "to a great extent."

Signaling

1. With respect to the target supplier, it can be said that they . . .
 - .. give us ample notice of planned price changes.
 - .. do a good job of notifying us in advance of any delivery schedule changes.
 - .. tell us of any changes in billing procedures well ahead of time.
 - .. would inform us early of any plans to modify the target product.
 - .. would discuss with us any plans to change the quality of the target product.
 - .. would give us plenty of notice if their level of after-sales service was going to change.

Items scored on 5-point scales ranging from "strongly disagree" to "strongly agree."

Frequency

1. How frequently did the target supplier use the following to communicate with your firm during the past year?
 - Letters or written reports?
 - Telephone calls?
 - Face-to-face meetings?

Items scored on 5-point scales labeled "6 months or less often," "Quarterly," "Bi-monthly," "Monthly," "Weekly or more often." Adapted from Van de Ven and Ferry (1980).

Richness

Computed as score for face-to-face item on frequency scale divided by frequency score.

Environmental Dynamism - Frequency Component

1. During the past year, how frequently have changes occurred in . . .
 - .. the product or service features desired by your customers?
 - .. the product or service features offered by your competitors?
 - .. the price sensitivity of your customers?
 - .. the pricing tactics of your competitors?

Items scored on 5-point scales ranging from "no changes" to "very frequent changes." Adapted from Achrol and Stern (1988) and Doty et al. (1988).

Environmental Dynamism - Effect Component

1. To what extent is your firm **affected by changes** in . . .
 - .. the product or service features desired by your customers?
 - .. the product or service features offered by your competitors?
 - .. the price sensitivity of your customers?
 - .. the pricing tactics of your competitors?

Items scored on 5-point scales ranging from "very little" to "very much." Adapted from Achrol and Stern (1988) and Doty et al. (1988).

Environmental Dynamism

Computed as the product of the frequency and effect components above.

Environmental Diversity - Consumers

1. How similar or different are your **individual** customers, in terms of . . .
 - .. demographic characteristics (income, profession, education, social class)?
 - .. preferences for product/service features?
 - .. preferences for after sales service?
 - .. credit needs?

Items scored on 5-point scales ranging from "very similar" to "very different." Adapted from Achrol and Stern (1988).

Environmental Diversity - Business Customers

1. How similar or different are your **business** customers, in terms of . . .
 - .. type of business?
 - .. preferences for product/service features?
 - .. preferences for after sales service?
 - .. credit needs?

Items scored on 5-point scales ranging from "very similar" to "very different." Adapted from Achrol and Stern (1988).

Environmental Diversity

Computed as average of consumer and business components, weighted by proportion of customer type.

Importance

1. How **important** is . . .

- . . the price of the target product to your firm's profitability?
- . . the quality of the target product to the quality of your products?
- . . on-time delivery for maintaining smooth operations in your firm?
- . . the level of after-sales service for maintaining smooth operations?

Items scored on 5-point scales ranging from "very unimportant" to "very important."

Age of the relationship

1. For **how many years** has your firm been buying from the **target supplier**?

Product Rating

1. Please rate the **target supplier's**:

- product quality
- price
- ability to meet delivery commitments

Items scored on 5-point scales ranging from "very unsatisfactory" to "very satisfactory."

Sellers' Dependence

1. Approximately what **percent** of the target supplier's **total** sales did your firm provide in the past year?

Buyers' percentage estimates were used as score for dependence.

Availability of Alternative Suppliers

1. Please rate the **availability** of alternative suppliers for the target product.

Item scored on 5-point scale ranging from "very few available" to "very many available."

APPENDIX B



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June 22, 1990

Mr. Norman P. Addington
Procurement Manager
Sperry Marine
1070 Seminole Trail
Charlottesville, VA 22901

Dear Mr. Addington:

I am writing to request your assistance in connection with my doctoral dissertation on organizational buyer-seller relationships. The purpose of this research is to (1) understand the relative importance of factors affecting buyer-seller relationships, and (2) suggest ways for enhancing such relationships, from the buyer's point of view.

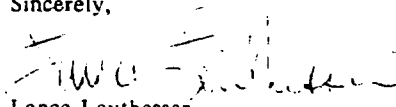
I would greatly appreciate your completing the enclosed questionnaire and returning it in the self-addressed envelope. In preliminary interviews, business people like yourself took about 15 minutes to fill out the questionnaire and found it very interesting. SINCE YOU ARE ONE OF A SMALL NUMBER OF N.A.P.M. MEMBERS BEING CONTACTED, YOUR RESPONSE IS VERY IMPORTANT FOR THE SUCCESSFUL COMPLETION OF THIS RESEARCH AND THE VALIDITY OF ITS FINDINGS. I would like to emphasize that your responses are confidential, and will be used strictly for research purposes.

I would be pleased to provide you with a summary of the research findings as soon as the study is completed. You may request a copy by providing your name and address on the questionnaire, or you may write to me separately to ensure anonymity of your responses.

I would be happy to answer any questions you might have. Please do not hesitate to call me at (512) 471-1128 (work) or (512) 338-1747 (home).

Thank you for your assistance.

Sincerely,


Lance Leuthesser
Doctoral Candidate

Last week I contacted you, requesting your assistance in connection with my doctoral research on buyer-seller relationships. If you have already completed and returned the study questionnaire, please accept my sincere thanks. If not, could I please ask you to do so today.

I am writing to you again because only a limited number of N.A.P.M. members are being contacted. Therefore, it is very important that each person contacted return the completed questionnaire. It will only take 15 minutes, and your responses are completely confidential.

I will greatly appreciate your cooperation.

Sincerely,

A handwritten signature in black ink, appearing to read "Lance Leuthesser". The signature is fluid and cursive, with the first name "Lance" written in a larger, more prominent script than the last name "Leuthesser".

Lance Leuthesser
Doctoral Candidate
The University of Texas at Austin



COLLEGE OF BUSINESS ADMINISTRATION
THE UNIVERSITY OF TEXAS AT AUSTIN

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July 13, 1990

Mr. Norman P. Addington
Procurement Manager
Sperry Marine
1070 Seminole Trail
Charlottesville, VA 22901

Dear Mr. Addington:

About three weeks ago I had written to you, requesting your participation in a survey on buyer-seller relationships. Although the response has been very encouraging, I have yet to receive some completed questionnaires. Because the responses are anonymous, I am uncertain if you have returned your questionnaire. If you have, please accept my sincere thanks.

If you have not yet returned the completed questionnaire, could you please take a few minutes to do so today? For the findings of this research to be valid, it is very important that each N.A.P.M. member in the scientifically selected sample respond to the survey. I would like to emphasize that your responses are strictly confidential.

I have enclosed a replacement questionnaire, in the event that the original one is misplaced. Also enclosed is an extra pre-paid return envelope. If you have any questions, please do not hesitate to call me at (512) 471-1128 (office) or (512) 338-1747 (home).

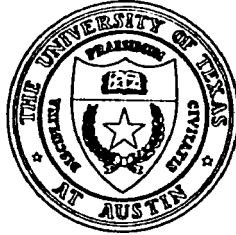
Thank you very much for your consideration.

Sincerely,

A handwritten signature in dark ink, appearing to read "Lance Leuthesser", written over a horizontal line.

Lance Leuthesser
Doctoral Candidate

APPENDIX C



University of Texas at Austin

SURVEY OF BUSINESS BUYERS

General Instructions

This questionnaire concerns your firm's relationship with one of your suppliers. If you belong to a large corporation, please think of your "firm" as being your business unit or division, not the entire corporation. If a question does not apply, or if you cannot answer it, please leave it blank.

Please mail the completed survey in the return envelope provided.

Thank you for your help!

IMPORTANT

Please take a moment to think of your firm's most recent product purchase meeting the following conditions:

- The product is at least moderately important to your firm.
- You purchase the product from two or more competing suppliers.
- You are very familiar with your firm's relationships with these suppliers.

Please select the supplier with the largest share of your business for the product. (In case of a tie, pick one of them).

SECTION I

The following questions pertain to the product and supplier you have just selected, referred to as the "target product" and "target supplier." If you belong to a large corporation, please keep in mind that your "firm" means your business unit or division.

1. The target product is best described as:

- () Capital or Accessory Equipment
- () Raw or Semi-finished Materials
- () Parts or Components
- () Operating or Maintenance Supplies

Other (please specify) _____

2. For how many years has your firm been buying from the target supplier?

_____ yrs.

3. In all, how many suppliers did you use for the target product during the past year?

4. What percent of last year's target product purchases (dollar value) were from the target supplier?

_____ %

5. Assuming you were not bound by contractual or other constraints (and the target supplier had no supply limitations), what share would the target supplier have received?

_____ %

6. Approximately what percent of the target supplier's total sales did your firm provide in the past year?

_____ %

7. What are your total annual purchases of the target product?

\$ _____

8. Over the past year, to what extent have your purchases from the target supplier changed, in terms of . . .
- | | Decreased
Greatly | | | | Increased
Greatly |
|-------------------------------|----------------------|---|---|---|----------------------|
| . . . dollar value? | 1 | 2 | 3 | 4 | 5 |
| . . . share of your business? | 1 | 2 | 3 | 4 | 5 |
9. How important is . . .
- | | Very
Unimportant | | | | Very
Important |
|---|---------------------|---|---|---|-------------------|
| . . . the price of the target product to your firm's profitability? | 1 | 2 | 3 | 4 | 5 |
| . . . the quality of the target product to the quality of your products? | 1 | 2 | 3 | 4 | 5 |
| . . . on-time delivery for maintaining smooth operations in your firm? | 1 | 2 | 3 | 4 | 5 |
| . . . the level of after-sales service for maintaining smooth operations? | 1 | 2 | 3 | 4 | 5 |
10. Please rate the target supplier's:
- | | Very
Unsatisfactory | | | | Very
Satisfactory |
|--|------------------------|---|---|---|----------------------|
| - product quality | 1 | 2 | 3 | 4 | 5 |
| - price | 1 | 2 | 3 | 4 | 5 |
| - ability to meet delivery commitments | 1 | 2 | 3 | 4 | 5 |
11. Please rate the availability of alternative suppliers for the target product.
- | | Very Few
Available | | | | Very Many
Available |
|--|-----------------------|---|---|---|------------------------|
| | 1 | 2 | 3 | 4 | 5 |

SECTION II

This section deals with your firm's relationship with the target supplier.

1. With respect to the target supplier, it can be said that they . . .
- | | Strongly
Disagree | | | | Strongly
Agree |
|---|----------------------|---|---|---|-------------------|
| . . . give us ample notice of planned price changes. | 1 | 2 | 3 | 4 | 5 |
| . . . do a good job of notifying us in advance of any delivery schedule changes. | 1 | 2 | 3 | 4 | 5 |
| . . . tell us of any changes in billing procedures well ahead of time. | 1 | 2 | 3 | 4 | 5 |
| . . . would inform us early of any plans to modify the target product. | 1 | 2 | 3 | 4 | 5 |
| . . . would discuss with us any plans to change the quality of the target product. | 1 | 2 | 3 | 4 | 5 |
| . . . would give us plenty of notice if their level of after-sales service was going to change. | 1 | 2 | 3 | 4 | 5 |

	Strongly Disagree				Strongly Agree
... would try to cover it up if they had a manufacturing setback.	1	2	3	4	5
... tend to be secretive about politics in their company.	1	2	3	4	5
... would freely admit it if they were having internal labor problems.	1	2	3	4	5
... tend to sidestep talk about weaknesses in their organization.	1	2	3	4	5
... are forthcoming in discussing managerial promotions and changes in their company.	1	2	3	4	5
... give us a clear picture of what goes on behind the scenes in their company.	1	2	3	4	5
2. Keeping in mind your firm's overall relationship with the target supplier, please indicate the <u>extent</u> to which they ...	Not at All				To a Great Extent
... spend time trying to learn how they can help you lower the cost of using the target product.	1	2	3	4	5
... spend time trying to help you gain maximum benefit from the target product.	1	2	3	4	5
... make an effort to keep abreast of your customers' changing needs.	1	2	3	4	5
... show an interest in helping you become more competitive.	1	2	3	4	5
... keep up with changes in your competitive environment.	1	2	3	4	5
... show interest in understanding how your firm works.	1	2	3	4	5
	Strongly Disagree				Strongly Agree
3. We are confident they would charge us fair prices even if they had an opportunity to take advantage of us.	1	2	3	4	5
4. We're afraid they would cut corners if we let up on inspection.	1	2	3	4	5
5. If we were facing unprofitable times, we believe that they would share the burden by trimming their margins.	1	2	3	4	5
6. If it was important to us, we could count on them to do a rush job even if they had to absorb some additional costs.	1	2	3	4	5
7. They readily acknowledge it if another supplier can meet our needs better.	1	2	3	4	5
8. We are confident in their ability to help us solve any problems involving the target product.	1	2	3	4	5
9. From our perspective, this supplier is very competent.	1	2	3	4	5
10. We are delighted with our overall relationship with them.	1	2	3	4	5
11. We wish more of our suppliers were like this one.	1	2	3	4	5

	Strongly Disagree			Strongly Agree		
12. We would like our relationship with them to continue in the coming years.	1	2	3	4	5	
13. It is a pleasure dealing with this supplier.	1	2	3	4	5	
14. There is always some problem or another with this supplier.	1	2	3	4	5	
15. There are close ties between us and this supplier.	1	2	3	4	5	

SECTION III

This section concerns the way your firm interacts with the target supplier.

- Please give 10 points to the department in your firm which communicated most frequently with the target supplier over the past year, then give 0 to 9 points to each of the remaining departments, per the following example:

Example

If your engineering department communicated the most with the target supplier over the past year, then mark "10" beside "R&D/Engineering." If your purchasing department communicated with the target supplier about half as frequently as engineering, then mark "5" beside "Purchasing/Materials Management." Follow the same procedure for each of the remaining departments.

- Sales/Marketing _____
- Production/Quality Assurance _____
- R&D/Engineering _____
- Finance/Accounting _____
- Transportation/Distribution _____
- Purchasing/Materials Management _____

- Please give 10 points to the level in your firm which communicated most frequently with the target supplier over the past year, and 0 to 9 points to the remaining levels, as above.

- Top Management (Directors, Pres., Exec. VP, and so on) _____
- Functional Vice President _____
- Department Manager (Sales Mgr., Purch. Mgr., and so on) _____
- Department Members (Sales Rep., Purch. Agent, and so on) _____
- Clerical _____

3. Now, please focus on the target supplier and give 10 points to the department in their company which communicated most frequently with your firm over the past year, and 0 to 9 points to the remaining departments.

- Sales/Marketing
- Production/Quality Assurance
- R&D/Engineering
- Finance/Accounting
- Transportation/Distribution
- Purchasing/Materials Management

4. Please give 10 points to the level in the target supplier which communicated most frequently with your firm over the past year, and 0 to 9 points to the remaining levels.

- Top Management (Directors, President, Exec VP, and so on)
- Functional Vice President
- Department Manager (Sales Mgr., Purch. Mgr., and so on)
- Department Members (Sales Rep., Purch. Agent, and so on)
- Clerical

5. How frequently did the target supplier use the following to communicate with your firm during the past year? (Please circle the best approximation.)

6 months or less often
Quarterly
Bi-monthly
Monthly
Weekly or more often

- Letters or written reports?
- Telephone calls?
- Face-to-face meetings?

1	2	3	4	5
1	2	3	4	5
1	2	3	4	5

SECTION IV

This section deals with your market (i.e., your customers and competitors). If your firm competes in several markets, select one of the more important ones. The selected market should be one where your performance is affected by the target product. (In certain instances, customers may include other business units or divisions within your corporation.)

1. During the past year, how <u>frequently</u> have changes occurred in . . .	No Change				Very Frequent Change
. . . the product or service features desired by your customers?	1	2	3	4	5
. . . the product or service features offered by your competitors?	1	2	3	4	5
. . . the price sensitivity of your customers?	1	2	3	4	5
. . . the pricing tactics of your competitors?	1	2	3	4	5
2. To what extent is your firm <u>affected by changes</u> in . . .	Very Little				Very Much
. . . the product or service features desired by your customers?	1	2	3	4	5
. . . the product or service features offered by your competitors?	1	2	3	4	5
. . . the price sensitivity of your customers?	1	2	3	4	5
. . . the pricing tactics of your competitors?	1	2	3	4	5
3. What percentage of your <u>immediate customers</u> are . . .					
. . . individual customers (consumers)?					%
. . . business customers (including gov't. agencies)?					%
Total			100		%
4. How similar or different are your <u>individual</u> customers, in terms of . .	Very Similar				Very Different
. . . demographic characteristics (income, profession, education, social class)?	1	2	3	4	5
. . . preferences for product/service features?	1	2	3	4	5
. . . preferences for after sales service?	1	2	3	4	5
. . . credit needs?	1	2	3	4	5
5. How similar or different are your <u>business</u> customers, in terms of					
. . . type of business?					
. . . preferences for product/service features?	1	2	3	4	5
. . . preferences for after sales service?	1	2	3	4	5
. . . credit needs?	1	2	3	4	5

SECTION V

Please provide the following general information.

1. Approximately how many employees are there in your firm? _____
2. Approximately how many employees are there in the target supplier's organization? _____
3. What are your firm's approximate annual sales (including intercompany transfers)? \$ _____
4. Your firm's primary product is best described as:
 - () Consumer products (durable or non-durable)
 - () Industrial equipment, parts, or components
 - () Raw materials or industrial supplies
 - () Services
 - () Retail/Wholesale
 - () Other (please specify) _____
5. If you would like to receive a report summary, please provide your name and address (if you prefer, you may request it under separate cover).
6. Your response to this question is optional. We would like the name and address of a person in another department in your firm, who is also familiar with the target product and target supplier, whom we may contact to request similar information.

THANK YOU VERY MUCH FOR YOUR TIME AND COOPERATION!

APPENDIX D

Chow Test - Example

- Step 1: Order observations on moderator variable of interest ("age" in the following example).
- Step 2: Subgroup by introducing 0,1 dummy variable (low 40% coded "0", high 40% coded "1").
- Step 3: Estimate "unconstrained" regression model, which includes (predictor variable) X (subgroup code) interaction term for each independent variable, per the following example:

$$\begin{aligned} \text{TRST} = & b_0 + b_1 \text{RATE} + b_2 \text{DISC} + b_3 \text{INIT} + b_4 \text{SIG} + b_5 \text{FREQ} + \\ & + b_6 \text{RICH} + b_7 \text{DUM} + b_8 \text{RATEX} + b_9 \text{DISCX} + b_{10} \text{INITX} \\ & + b_{11} \text{SIGX} + b_{12} \text{FREQX} + b_{13} \text{RICHX} \end{aligned}$$

where:

TRST	=	Trust
RATE	=	Product rating
DISC	=	Disclosing
INIT	=	Initiating
SIG	=	Signaling
FREQ	=	Frequency
RICH	=	Richness
DUM	=	Dummy variable = "0" for low age, "1" for high age
RATEX	=	Product rating X dummy
DISCX	=	Disclosing X dummy
INITX	=	Initiating X dummy
SIGX	=	Signaling X dummy
FREQX	=	Frequency X dummy
RICHX	=	Richness X dummy

- Step 4: Estimate "constrained" model by omitting interaction term of independent variable of interest ("frequency" in this example):

$$\begin{aligned} \text{TRST} = & b_0 + b_1 \text{RATE} + b_2 \text{DISC} + b_3 \text{INIT} + b_4 \text{SIG} + b_5 \text{FREQ} \\ & + b_6 \text{RICH} + b_7 \text{DUM} + b_8 \text{RATEX} + b_9 \text{DISCX} + b_{10} \text{INITX} \\ & + b_{11} \text{SIGX} + b_{12} \text{RICHX} \end{aligned}$$

Step 5: Compute test statistic as follows:

$$T = \frac{SSE_c - SSE_u}{(SSE_u) / (N - p)}$$

where:

T = Test statistic
SSE_c = Error sum of squares, constrained model
SSE_u = Error sum of squares, unconstrained model
N = Number of observations
p = Number of parameters estimated in unconstrained model

Step 6: Compare T to $F_{crit} = F_{1, N-p, .05}$

REFERENCES

- Achrol, Ravi and Louis Stern (1988), "Environmental Determinants of Decision-Making Uncertainty in Marketing Channels," Journal of Marketing Research, 25(February), 26-50.
- Ancona, Deborah G. and David F. Caldwell (1988), "Beyond Task and Maintenance," Group and Organization Studies, 13(4), 468-493.
- and Barton Weitz (1989), "Determinants of Continuity in Conventional Industrial Channel Dyads", Marketing Science, 8(4), 310-323.
- Anderson, James C. and James A. Narus (1990), "A Model of Distributor Firm and Manufacturer Firm Working Relationships," Journal of Marketing, 54(January), 42-58.
- and James A. Narus (1984), "A Model of the Distributor's Perspective of Distributor-Manufacturer Working Relationships", Journal of Marketing, 48(Fall), 62-74.
- Arndt, Johan (1979), "Toward a Concept of Domesticated Markets," Journal of Marketing, 43(Fall), 69-75.
- Arnold, Hugh J. (1982), "Moderator Variables: A Clarification of Conceptual, Analytic, and Psychometric Issues," Organizational Behavior and Human Performance, 29, 143-174.
- Berry, Leonard L. (1980), "Services Marketing is Different," Business, 30(May-June), 24-29.
- Blau, Peter M. (1974), On the Nature of Organizations, New York, NY: John Wiley and Sons.
- Bradford, Leland P. (1964), "Membership and the Learning Process," in T-Group Theory and Laboratory Method, Leland P. Bradford, Jack R. Gibb and Kenneth D. Benne (eds.), NY: John Wiley & Sons, 190-215.

- Campbell, Donald T. and Donald W. Fiske (1959), "Convergent and Discriminant Validation by the Multitrait-Multimethod Matrix," Psychological Bulletin, 56(2), 81-105.
- Carmen, James M. and Eric Langeard (1980), "Growth Strategies of Service Firms," Strategic Management Journal, 1(January-March), 7-22.
- Carnevale, P. J. D., Dean G. Pruitt and S. Seilheimer (1981), "Looking and Competing: Accountability and Visual Access in Integrative Bargaining," Journal of Personality and Social Psychology, 40, 111-120.
- Chow, Gregory C. (1960), "Test of Equality Between Sets of Coefficients in Two Linear Regressions," Econometrika, 28(July), 591-605.
- Daft, Richard L. and Robert H. Lengel (1986), "Organizational Information Requirements, Media Richness and Structural Design," Management Science, 32(5), 554-571.
- Day, Ralph L., Ronald E. Michaels and Barbara C. Perdue (1988), "How Buyers Handle Conflict," Industrial Marketing Management, 17, 153-160.
- Deutsch, Morton (1958), "Trust and Suspicion," Journal of Conflict Resolution, 2(4), 265-279.
- Dimock, Hedley G. (1986), Groups: Leadership and Group Development, San Diego, CA: University Associates, Inc.
- Dion, Paul A. and Peter M. Banting (1988), "Industrial Supplier-Buyer Negotiations," Industrial Marketing Management, 17, 43-47.
- Doty, D. H., W. H. Glick, K. M. Sutcliffe, C. C. Miller and G. P. Huber (1988), Code Study Technical Report 06, Department of Management, University of Texas at Austin.
- Duncan, Robert B. (1972), "Characteristics of Organizational Environments and Perceived Environmental Uncertainty," Administrative Science Quarterly, 17, 313-327.
- Dwyer, F. Robert and Rosemary R. Lagace (1986), "On the Nature and Role of Buyer-Seller Trust," AMA Proceedings, 40-45.

- and Sejo Oh (1987), "Output Sector Munificence Effects on the Internal Political Economy of Marketing Channels," Journal of Marketing Research, 24(November), 347-358.
- , Paul H. Schurr and Sejo Oh (1987), "Developing Buyer-Seller Relationships," Journal of Marketing, 51(April), 11-27.
- and M. Ann Welsh (1985), "Environmental Relationships of the Internal Political Economy of Marketing Channels", Journal of Marketing Research, 22(November), 397-414.
- Egan, Gerard (1976), Interpersonal Living: A Skills/Contract Approach to Human Relations Training in Groups, Monterey, CA: Brooks/Cole Publishing Co.
- Farace, Richard, Peter Monge and Hamish Russell (1977), Communication and Organizing, Reading, MA: Addison-Wesley Publishing Co.
- Frazier, Gary L., Robert E. Spekman and Charles R. O'Neal (1988), "Just-in-time Exchange Relationships in Industrial Markets," Journal of Marketing, 52(October), 52-67.
- Gibb, Jack R. (1964), "Climate for Trust Formation," in T-Group Theory and Laboratory Method, Leland P. Bradford, Jack R. Gibb and Kenneth D. Benne (eds.), NY: John Wiley & Sons, 279-309.
- Giffen, Kim (1967), "The Contribution of Studies of Source Credibility to Theory of Interpersonal Trust in the Communication Process," Psychological Bulletin, 68(2), 104-120.
- Golembiewski, Robert T. and Mark McConkie (1975), "The Centrality of Interpersonal Trust in Group Processes," in Theories of Group Processes, Cary L. Cooper (ed.), NY: John Wiley & Sons, 131-185.
- Gronroos, Christian (1978), "A Service-Oriented Approach to Marketing," European Journal of Marketing, 12(8), 588-601.
- Guadagni, Peter M. and John D. C. Little (1983), "A Logit Model of Brand Choice Calibrated on Scanner Data," Marketing Science, 2(3), 203-238.

- Huber, George P. and Richard L. Daft (1987), "The Information Environments of Organizations," in Handbook of Organizational Communication, Beverly Hills, CA: Sage Publications.
- and Thomas W. Speh (1989), Business Marketing Management, Chicago: The Dryden Press.
- Johnson, David W. and Frank P. Johnson (1987), Joining Together: Group Theory and Group Skills, Englewood Cliffs, NJ: Prentice-Hall, Inc.
- Johnston, Wesley J. and Thomas V. Bonoma (1981), "The Buying Center: Structure and Interaction Patterns," Journal of Marketing, 45(Summer), 143-156.
- Katz, Daniel and Robert L. Kahn (1966), The Social Psychology of Organizations, New York, NY: John Wiley and Sons.
- Lattin, James M. and Randolph E. Bucklin (1989), "Reference Effects of Price and Promotion on Brand Choice Behavior," Journal of Marketing Research, 26(3), 299-310.
- Leblebici, Huseyin and Gerald R. Salancik (1981), "Effects of Environmental Uncertainty on Information and Decision Processes in Banks," Administrative Science Quarterly, 26(December), 578-596.
- Leifer, Richard and George P. Huber (1977), "Relations among Perceived Environmental Uncertainty, Organization Structure, and Boundary-Spanning Behavior," Administrative Science Quarterly, 22, 235-247.
- Levitt, Theodore (1983), "After the Sale is Over", Harvard Business Review, 61(September-October), 87-93.
- Loomis, J. L. (1959), "Communication, the Development of Trust and Cooperative Behavior," Human Relations, 12(4), 305-315.
- Lovelock, Christopher H. (1981), "Why Marketing Management Needs to be Different for Services," in Marketing of Services, J. H. Donnelly and W. R. George (eds.), Chicago, IL: American Marketing Association.

- Lysonski, Steven (1985), "A Boundary Theory Investigation of the Product Manager's Role," Journal of Marketing, 49(Winter), 26-40.
- Macauley, Stewart (1963), "Non-contractual Relations in Business: A Preliminary Study," American Sociological Review, 28(1), 55-69.
- Macneil, Ian R. (1980), The New Social Contract: An Inquiry into Modern Contractual Relations, New Haven, CT: Yale University Press.
- March, James G. and Zur Shapira (1987), "Managerial Perspectives on Risk and Risk Taking," Management Science, 33(11), 1404-1418.
- Martin, R. J. (1983), A Skills and Strategies Handbook for Working with People, Englewood Cliffs, NJ: Prentice-Hall.
- McCann, John and Jay R. Galbraith (1981), "Interdepartmental Relations," Handbook of Organizational Design, Vol. 2, Paul C. Nystrom and William Starbuck (eds.), New York, NY: Oxford University Press, 60-84.
- McQuiston, Daniel (1989), "Novelty, Complexity, and Importance as Causal Determinants of Industrial Buyer Behavior," Journal of Marketing, 53(April), 66-79.
- Miles, Raymond E. and Charles C. Snow (1978), Organizational Strategy, Structure and Process, New York, NY: McGraw-Hill Book Co.
- Neter, John, William Wasserman and Michael H. Kutner (1983), Applied Linear Regression Models, Homewood, IL: Richard D. Irwin.
- Perdue, Barbara C., Ralph L. Day and Ronald E. Michaels (1986), "Negotiation Styles of Industrial Buyers," Industrial Marketing Management, 15, 171-176.
- Phillips, Lynn W. (1981), "Assessing Measurement Error in Key Informant Reports: A Methodological Note on Organizational Analysis in Marketing," Journal of Marketing Research, 28(November), 395-415.
- Pruitt, Dean G. (1981), Negotiation Behavior, NY: Academic Press.

- , and Steven A. Lewis (1975), "Development of Integrative Solutions in Bilateral Negotiations," Journal of Personality and Social Psychology, 31, 621-633.
- , and Steven A. Lewis (1977), "The Psychology of Integrative Bargaining," in Negotiations: Social Psychological Perspectives, Daniel Druckman (ed.), Beverly Hills, CA: Sage Publications.
- Rempel, John K., John G. Holmes, and Mark P. Zanna (1985), "Trust in Close Relationships," Journal of Personality and Social Psychology, 49(1), 95-112.
- Rotter, Julian B. (1971), "Generalized Expectancies for Interpersonal Trust," American Psychologist, 26(5), 443-452.
- Ruekert, Robert W. and Orville C. Walker Jr. (1987), "Marketing's Interaction with Other Functional Units: A Conceptual Framework and Empirical Evidence," Journal of Marketing, 51(January), 1-19.
- , Orville C. Walker Jr., and Kenneth J. Roering (1985), "The Organization of Marketing Activities: A Contingency Theory of Structure and Performance," Journal of Marketing, 49(Winter), 13-25.
- Schopler, Janice H. (1987), "Interorganizational Groups: Origins, Structure, and Outcomes," Academy of Management Review, 12(4), 702-713.
- Schurr, Paul H. and Julie L. Ozanne (1985), "Influences on Exchange Processes: Buyers' Perceptions of a Seller's Trustworthiness and Bargaining Toughness," Journal of Consumer Research, 11(March), 939-953.
- Schwab, Robert C., Gerardo R. Ungson and Warren B. Brown (1985), "Redefining the Boundary Spanning - Environment Relationship," Journal of Management, 11(1), 75-86.
- Sharma, Subash, Richard M. Durand and Oded Gur-Arie (1981), "Identification and Analysis of Moderator Variables," Journal of Marketing Research, 18(August), 291-300.
- Shaw, Marvin E. (1981), Group Dynamics: The Psychology of Small Group Behavior, NY: McGraw-Hill Book Co.

- Shostack, G. Lynn (1977), "Breaking Free from Product Marketing," Journal of Marketing, 41(April), 73-80.
- Thompson, James D. (1967), Organizations in Action, New York, NY: McGraw-Hill Book Company.
- Van de Ven, Andrew H. (1976), "On the Nature, Formation, and Maintenance of Relations Among Organizations", Academy of Management Review, (October), 24-36.
- , and Diane L. Ferry (1980), Measuring and Assessing Organizations, New York, NY: John Wiley and Sons.
- , and Gordon Walker (1984), "The Dynamics of Interorganizational Coordination," Administrative Science Quarterly, 29, 598-621.
- Walker, Orville C. Jr. and Robert W. Ruekert (1987), "Marketing's Role in the Implementation of Business Strategies: A Critical Review and Conceptual Framework," Journal of Marketing, 51(July), 15-33.
- Weick, Karl (1979), The Social Psychology of Organizing, Reading, MA: Addison-Wesley Publishing Co.
- Williamson, Oliver E. (1979), "Transaction-Cost Economics: The Governance of Contractual Relations," Journal of Law and Economics, 22(2), 233-261.
- Zeithaml, Valerie A., A. Parasuraman and Leonard L. Berry (1985), "Problems and Strategies in Services Marketing," Journal of Marketing, 49(Spring), 33-46.

VITA

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